THE JOURNAL OF COMPARATIVE NEUROLOGY 421:29–51 (2000)

Somatosensory and Motor Representations in Cerebral Cortex of a Primitive Mammal (*Monodelphis domestica*): A Window Into the Early Evolution of Sensorimotor Cortex

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ABSTRACT

To examine the potential early stages in the evolution of sensorimotor cortex, electrophysiological studies were conducted in the primitive South American marsupial opossum, *Monodelphis domestica.* Somatosensory maps derived from multiunit microelectrode recordings revealed a complete somatosensory representation of the contralateral body surface within a large region of midrostral cortex (primary somatosensory cortex, or S1). A large proportion (~51%) of S1 was devoted to representation of the glaborous snout, mystacial vibrissae, lower jaw, and oral cavity (the rostrum). A second representation, the second somatosensory area (or S2), was found adjacent and caudolateral to S1 as a mirror image reversed along the representation of the glabrous snout. A reversal of somatotopic order and an enlargement of receptive fields marked the transition from S1 to S2. Mapping of excitable cortex was conducted by using intracortical microstimulation (ICMS) techniques, as well as low-impedance depth stimulation and bipolar surface stimulation. In all three procedures, electrical stimulation resulted in movements confined strictly to the face. Specifically, at virtually all sites from which movements could be evoked, stimulation resulted in only vibrissae movement. ICMS-evoked vibrissae movements typically occurred at sites within S1 with receptive fields of the mystacial vibrissae, lower jaw, and glaborous snout. Results were similar using low-impedance depth stimulation and bipolar surface stimulation techniques except that the motor response maps were generally larger in area. There was no evidence of a motor representation rostral to S1. Examination of the cytoarchitecture in this cortical region (reminiscent of typical mammalian somatosensory cortex) and the high

This study was a collaborative effort between the laboratories of Bruce Masterton and Randolph Nudo. The manuscript was in preparation at the time of Bruce's death in 1996. This work would not have been possible without Bruce's unfailing guidance, wisdom and inspiration.

Grant sponsor: NIH-NINDS; Grant numbers: NS30853, NS07726; Grant sponsor: NICHD; Grant number: HD02528; Grant sponsor: The Kansas Claude Pepper Center for Independence in Older Americans; Grant number: NIA AG 14635; Grant sponsor: NRSA; Grant number: NS 09366.

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Received 24 August 1999; Revised 23 December 1999; Accepted 11 January 2000

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levels of stimulation needed for vibrissae movement suggest that the parietal neocortex of *Monodelphis* is representative of a primitive sensorimotor condition. It possesses a complete S1 representation with an incomplete motor component overlapping the S1 representation of the face. It contains no primary motor representation. Completion of the motor representations within S1 (trunk, limbs, tail) as well as the emergence of a primary motor cortex rostral to S1 may have occurred relatively late in mammalian phylogeny. J. Comp. Neurol. 421: 29–51, 2000. © 2000 Wiley-Liss, Inc.

Indexing terms: sensory cortex; motor cortex; marsupial; evolution

For at least 50 years, it has been suggested that the evolution of the mammalian sensorimotor cortex was marked by progressive differentiation into separate sensory and motor areas. This notion gained widespread support after a series of electrophysiological experiments by Richard Lende in the 1960s using surface stimulation and evoked potential techniques in the North American opossum (*Didelphis virginiana*). Based on these experiments, Lende argued that the sensory and motor areas of neurologically primitive mammals (e.g., opossums) overlap completely. He referred to this area as a "sensorimotor amalgam" (Lende, 1963a,b,c).

Recently, we reexamined the notion of a sensorimotor amalgam in the gray, short-tailed (or Brazilian) opossum (*Monodelphis domestica*). This species was chosen because in a previous comparative study of 24 mammals, it was found to possess relatively few corticospinal neurons, possibly indicating that its cortical motor system is neurologically primitive (Nudo and Masterton, 1988, 1990a). Of the species examined in tract tracing studies by Nudo and Masterton, the *Monodelphis* opossum was found to have the second lowest number of corticospinal neurons (second only to the least shrew), the lowest number of corticospinal neurons per brain weight, and the smallest proportion of cortex containing corticospinal neurons (Nudo and Masterton, 1990a,b; Nudo et al., 1995).

M. domestica is a small terrestrial marsupial native to the semiarid region of northeastern Brazil (Strilein, 1982). Like the North American opossum, *Monodelphis* is part of the family of Didelphid opossums which, based on comparative neurology of the past century, has retained the basic brain structure that is believed to closely approximate the form which can serve as a starting point for tracing the evolution of modern mammalian neural organization, whether marsupial or placental (Smith, 1910; Loo, 1930; Edinger, 1948; Simpson, 1949, 1959; Olsen, 1959; Frost and Masterton, 1992). This view has further been supported by fossil and geological evidence showing the existence of mammals virtually identical in bony structure to extant Didelphids at least since the Cretaceous and perhaps the Jurassic, 100 million or more years ago (Gregory, 1929; Clemens, 1968, 1977; Marshall et al., 1987). Furthermore, certain Monodelphid anatomical characteristics, such as small body size and lack of a marsupial pouch, suggest that this species may be more "primitive" and characteristic of the original metatherian form than *Didelphis virginiana* (Kirsch, 1977; Marshall, 1979; Reig et al., 1987).

Abbreviations

AUD	auditory cortex
C	caudal
contra	contralateral
D	dorsal
D1–D5	digits
distal fl.	distal forelimb
HL, hl	hindlimb
ICMS	intracortical microstimulation
ipsi	ipsilateral
l. back	lower back
l. lip	lower lip
l. jaw	lower jaw
M	medial
mid	middle
Ms	motorsensory cortex
Mv	motor vibrissae
NR	no response
R	rostral
Rh. Fiss.	rhinal fissure
S1	primary somatosensory cortex
S2	secondary somatosensory cortex
sh	shoulder
Sd	sensory digit
Sm	sensorymotor cortex
Sv	sensory vibrissae
Tot. SS	total somatosensory area
u. lip	upper lip
V	ventral
vb, vib	vibrissae
VIS	visual cortex

MATERIALS AND METHODS

Somatosensory and motor mapping procedures were conducted in 12 adult Gray Short-tailed opossums (*Monodelphis domestica;* 7 male, 5 female). Body weights ranged from 45 to 139 g. Combined somatosensory and motor mapping procedures were conducted in seven animals. In addition, exclusive somatosensory mapping (i.e., without motor mapping procedures) was conducted in one animal, whereas exclusive motor mapping procedures (i.e., without somatosensory procedures) were conducted in four animals. Three additional animals were used to examine the cortical architecture in sections from flattened cortex processed for myelin.

Surgical and electrophysiological techniques

Surgical techniques. First, each animal was preanesthetized to effect with halothane/nitrous oxide. Then one of three anesthetic protocols was followed during surgical and electrophysiological procedures, depending upon the particular requirements of the mapping experiment. For each anesthetic protocol, an initial dose was administered to establish a deep anesthetic level, then subsequent smaller doses were administered to maintain adequate anesthesia. As the anesthetics were given to effect, and varied across animals, the doses are approximate. These

anesthetic protocols were: (1) ketamine/xylazine for combined motor and somatosensory mapping procedures (n = 2) and for motor mapping procedures alone (n = 4): stock ketamine (100 mg/ml) and xylazine (20 mg/ml) were diluted with saline to achieve a concentration of 50 mg/ml and 5 mg/ml, respectively. Animals were given an initial intraperitoneal dose of 10 mg ketamine and 1 mg Xylazine per 100 g of body weight. Smaller doses (ketamine 5 mg/100 g, Xylazine .5 mg/100 g) were delivered every 20–40 minutes for maintenance of a surgical level of anesthesia; (2) sodium pentobarbital for sensory mapping procedures alone (n = 1): stock sodium pentobarbital (65 mg/ml) was diluted 1:1 with saline to a concentration of 32.5 mg/ml. Animals were given an initial dose of 4 mg/100 g body weight. Maintenance doses of 3 mg/100 g were delivered every 2–3 hours. In several animals in which combined sensory and motor mapping procedures were employed, sodium pentobarbital was used for sensory mapping procedures and ketamine/xylazine was used for the motor mapping procedures (n = 3); (3) urethane for sensory and motor mapping procedures (n = 2; 15% W/V in distilled water): animals were given an initial dose of 75 mg/100 g body weight. Maintenance doses of urethane were not required. The rationale for the use of urethane was for more direct comparison with an earlier study of somatosensory/motor overlap in Virginia opossums (Lende, 1963a,b,c). Neither the somatosensory nor the motor results differed qualitatively by using the different anesthetic protocols. Each animal was mounted in a stereotaxic frame, a laminectomy was performed between atlas and axis, and cerebrospinal fluid was allowed to drain to prevent swelling of the brain. An extensive craniotomy was performed over one hemisphere. A dental acrylic dam was constructed around the craniotomy, the dura was removed, and the brain was covered with a pool of warm silicone oil. A magnified photograph of the cortical surface vasculature was obtained by using a video frame grabber. Following the surgical procedure, somatosensory mapping procedures, motor mapping procedures, or both were conducted for a period of 1 to 18 hours. Core temperature was monitored and maintained by using a homeothermic blanket system within 32°C and 34°C.

All animals were cared for in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals and with the approval of the Institutional Animal Care and Use Committee.

Somatosensory mapping procedures. For multiunit somatosensory recordings, a glass micropipette filled with 3.5 M NaCl (impedance = 500 kΩ–1 MΩ), or a tungsten microelectrode (impedance = 1 MΩ) was introduced on a fine grid pattern, sited with reference to the surface vasculature (approximately 250–500 μm interpenetration distances; Fig. 1). Results obtained with the two microelectrode types were identical and are pooled in the remainder of the paper. At each site, the electrode was advanced perpendicular to the cortical surface to depths ranging from 400 to 800 μm. Minimal cutaneous receptive fields were defined by determining the skin field over which cortical neurons were driven by just visible indentation of the skin with a fine glass probe, or by hair bending. Somatosensory mapping procedures lasted 4 to 12 hours.

Motor mapping procedures. Motor mapping results were based primarily on intracortical microstimulation (ICMS) experiments. At each site, ICMS mapping procedures were employed to determine the movement(s) evoked by stimulation at near-threshold current levels. Details of this procedure are presented in Nudo et al. (1992, 1996). Briefly, at each site, a glass micropipette filled with 3.5 M NaCl (impedance = 500 kΩ–1 MΩ; outside tip diameter = 10–20 μm) was introduced on the same grid pattern used for somatosensory mapping. The electrode was advanced to approximately 1,200 μm, because movements were evoked at the lowest current levels at this depth. If motor responses were not evoked at this depth, other depths ranging from 800 to 1,700 μm were explored. This range corresponds approximately to layers 4 through 6. The stimulus consisted of 40-msec current trains of 200-μsec cathodal pulses delivered at 350 pulses/sec from an electrically isolated stimulation circuit. The maximum current used in these experiments was 60 μA, though higher currents were used at the conclusion of each experiment (up to 100 μA) to determine whether responses could be evoked at previously nonresponsive sites (n = 6 animals). In several cases, the train length of stimulation at sites where no movement could be evoked was extended to 80 ms.

Two additional stimulation procedures were used in four animals to determine whether any additional motor responses could be evoked by using techniques designed to stimulate larger volumes of cortical tissue, and hence, larger numbers of cortical neurons: 1) depth stimulation with low-impedance microelectrodes (hereafter called "low-impedance depth stimulation") employed electrodes with impedances of 100 kΩ. These electrodes also were glass micropipettes filled with 3.5 M NaCl, but outside tip diameters were larger. The use of these higher impedance electrodes required significantly higher current levels (maximum = 500 μA) to evoke movements (n = 3); 2) bipolar electrical stimulation of the cortical surface employed two fine stainless steel wires with a separation of 350–500 μm. These bipolar electrodes were similar to those used by Lende (1963c) and allow for a more direct comparison with his earlier study. Maximum current using bipolar stimulation was 3 mA (n = 4). Two of the animals used in bipolar surface stimulation mapping were also used in low-impedance depth stimulation mapping. Motor mapping procedures lasted 2 to 12 hours.

At the conclusion of the mapping procedures, reference marks were placed at the topographic boundaries within the sensorimotor area either by small electrolytic lesions or by injection of fluorescent latex microspheres.

Histology

Following electrophysiological procedures, the animals were deeply anesthetized with sodium pentobarbital (8 mg/100 g) and perfused transcardially with 0.9% saline followed by 10% formalin as fixative. The brains were then transferred to a solution of 30% sucrose formalin for 1–2 weeks. Coronal sections of 50-μm thickness were cut on a freezing microtome and mounted onto gelatin-coated slides and alternate sections stained for Nissl substance with cresyl violet. These sections were used to examine the cytoarchitecture and to determine the topographic location and depth of cortical recordings.

Three additional animals that had not undergone neurophysiologic procedures were anesthetized and perfused in the same manner, and their brains removed and stored in 5% formalin in 0.1 M phosphate buffer at pH 7.4 for 1–8 weeks. The cortex from each hemisphere was separated

from the brainstem and the underlying white matter, unfolded and flattened. Flattened cortices were stored 1–3 days between weighted glass slides in the same 5% buffered formalin. The cortex was then sectioned parallel to the surface at 30–50 μm on a freezing microtome. Sections were stored in fresh 5% phosphate-buffered formalin (pH 7.4) for 1–3 days and processed for myelin (Gallyas, 1979). These sections were examined to aid in the determination of general architectonic boundaries of somatosensory cortex in this species.

RESULTS

Detailed somatosensory maps were derived in five animals, ICMS motor maps in five animals, bipolar stimulation motor maps in four animals, and low-impedance depth stimulation motor maps in three animals.

Microelectrode mapping of multi-unit-evoked responses revealed a complete representation of the contralateral body surface within a large region of midrostral neocortex. In addition, a second, virtually complete, but smaller representation was revealed adjacent and lateral to the larger representation. The second representation was a mirror image of the first representation reversed along the representation of the glabrous snout.

Microelectrode electrical stimulation of this same region of neocortex resulted in movements that were confined strictly to the face. At virtually all sites from which movements could be evoked, electrical stimulation resulted only in vibrissae movement. These movements were found to roughly correspond to body parts congruent with the receptive fields recorded at each site. Detailed results of somatosensory and motor mapping experiments are reported below.

Microelectrode mapping of somatosensory cortex

Microelectrode recordings provided sufficient evidence for two somatotopically organized areas of cortex, one located rostromedially and the other caudolaterally, in four of the five opossums in which somatosensory mapping was completed (see Figs. 1, 2; and list of abbreviations). In the remaining opossum, the spatial resolution of the electrode penetrations was not sufficient to differentiate the two regions. The relative locations of electrode penetrations where neurons were responsive to light touch on the body surface or to movement of body hairs are shown for 5 cases in Figures 1 and 2. Area boundaries were estimated by drawing borders midway between two adjacent recording sites determined to be activated by cutaneous stimulation of different body parts. Sites with receptive fields of two or more body parts were often found on the border between the single body part representations. In each case, additional recording sites in adjacent regions of cortex were not activated by cutaneous stimuli, and thus were defined as "nonresponsive." Boundaries between the two somatotopically organized body representations were estimated by a reversal in somatotopic order and an increase in receptive field size (as described in detail below) and are delineated by a dashed line.

Primary somatosensory area (S1). Somatosensory mapping procedures revealed a complete representation of the contralateral body surface in which the hindlimb and tail were represented medially and the face and snout were represented laterally. This medial-to-lateral sequence is similar to the location and somatotopic order of S1 as observed in other mammals (Kaas, 1983). Figure 1 shows the topographic location of somatosensory recording sites in case 9213 and their location relative to previously identified corticospinal somata (Fig. 1, inset; Nudo and Masterton, 1990a). In this and other cases, the representation of the hindlimb and tail was located at or near the cortical midline. In the majority of cases the primary somatosensory area contained a relatively large representation of the distal forelimb lateral to the trunk and hindlimb representations, and a large representation of the face and snout, particularly the vibrissae, lateral to the forelimb representation. Figure 2 shows the topographic representation of the body surface in four other animals. In each case the somatotopic representation was located in a similar medial to lateral and rostral to caudal position in cortex.

The vast majority of responses were activated by stimulation of the contralateral side of the body. The only exceptions were at some sites activated by stimulation of the glabrous snout, midline face, lower jaw (near the midline), and back. Receptive fields at these sites were often bilateral.

Figure 3 shows typical receptive fields from representative sites in case 9213. In general, the multi-unit receptive fields of S1 in *Monodelphis* were larger in the size of the body part representation than those seen in primates using similar techniques (Merzenich et al., 1978; Sur et al., 1982). Many receptive fields of the hand extended over multiple digits, but were usually confined to either the glabrous or dorsal surface, but not both. Receptive fields of the vibrissae typically included several vibrissae and often entire rows. Receptive fields of the body were often quite large, covering the entire trunk and abdomen.

In cases with sufficient detail, a crudely somatotopic representation of the distal forelimb was consistently observed. In these cases, the ulnar digits of the distal forelimb were represented medially and the radial digits were represented laterally (Fig. 4). Some sites had receptive fields confined to single digits. An average of 23% of S1 was devoted to the distal forelimb (see Table 1). In addition, a crude caudal to rostral somatotopic gradient was found. That is, wrist and forearm receptive fields were always located in the caudal half of the forelimb representation, whereas digit fields were more rostral. This arrangement is similar to the topography of the forelimb representation in other species (see Kaas, 1983 for review).

Somatotopy was also evident in the face representation across S1 (Fig. 5). First, the lower jaw representation was typically located rostromedial to the upper jaw representation (Figs. 5, 6). Second, proceeding in a caudomedial to rostrolateral direction within the upper or lower jaw representation, receptive fields were found first on the posterior aspect of the jaw and then on progressively more anterior aspects of the jaw and/or snout and oral cavity (lower jaw: sites A, and J through M in Fig. 5; upper jaw: sites 1 through 6 and C in Fig. 6). Third, a crude somatotopic arrangement was observed for the mystacial vibrissae. That is, receptive fields for sites progressing in a medial to lateral direction indicated that the posterior mystacial vibrissae were represented medial to anterior mystacial vibrissae and snout (sites B–D in Fig. 5 and sites 1–4 in Fig. 6). A more orderly somatotopic arrangement of specific single vibrissae or rows

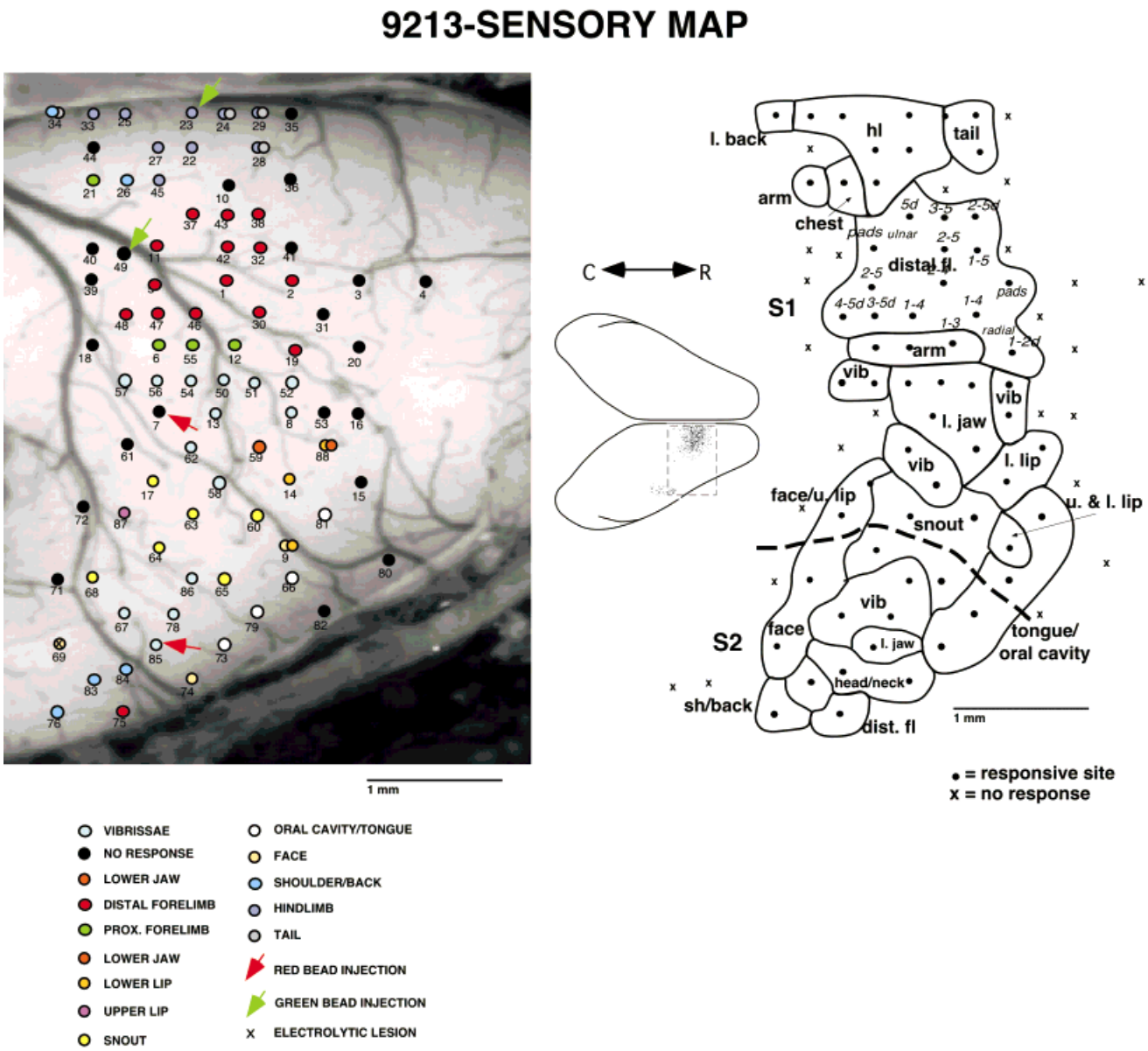


Fig. 1. Scope of somatosensory mapping experiments. Mapping procedures were guided by the locations of corticospinal neurons (**inset**) derived in previous experiments (Nudo and Masterton, 1990). In each animal, either somatosensory or motor mapping procedures, or both, were conducted. Left, dorsolateral photomicrograph of the right cortical hemisphere of *Monodelphis* 9213 showing the extent of the electrophysiological sensory recording. Right, the somatosensory map revealed a complete representation of the body surface (primary somatosensory area, S1), dominated by a large representation of the face, including the snout and vibrissae. As in other mammals, the hindlimb and tail are represented medially and the face is represented laterally. The distal forelimb comprised a significant portion of the map and a crude somatotopic organization was found for digits, with radial digits represented laterally and ulnar digits represented medially. A second representation (secondary somatosensory area, S2) of the body surface with higher thresholds and larger receptive fields was found lateral and caudal to S1. An electrolytic lesion and fluorescent bead injections were placed within the map as orientation markers for histology. Figure 1–14 were produced by using Canvas 3.5.5. For abbreviations in this and subsequent figures, see list.

of vibrissae was sought, but not found. The size of the vibrissae representation ranged from 12% to 18% of S1 (see Table 1). Half (51.3 ± 6.7%) of the somatosensory representation was composed of sites activated by stimulation of the rostral face: the glabrous snout, mystacial vibrissae, lower jaw, lips, and oral cavity (referred to as "rostrum" in Table 1). Finally, a representation of the lips, tongue, and oral cavity was consistently found occupying an area rostral to the snout representation. The trunk, hindlimb, and tail were represented in the medial portion of the map, with the tail represented rostral to the hindlimb and lower back representations (Fig. 1). The posterior body representation typically had large receptive fields and averaged 17.7% of the total S1 area.

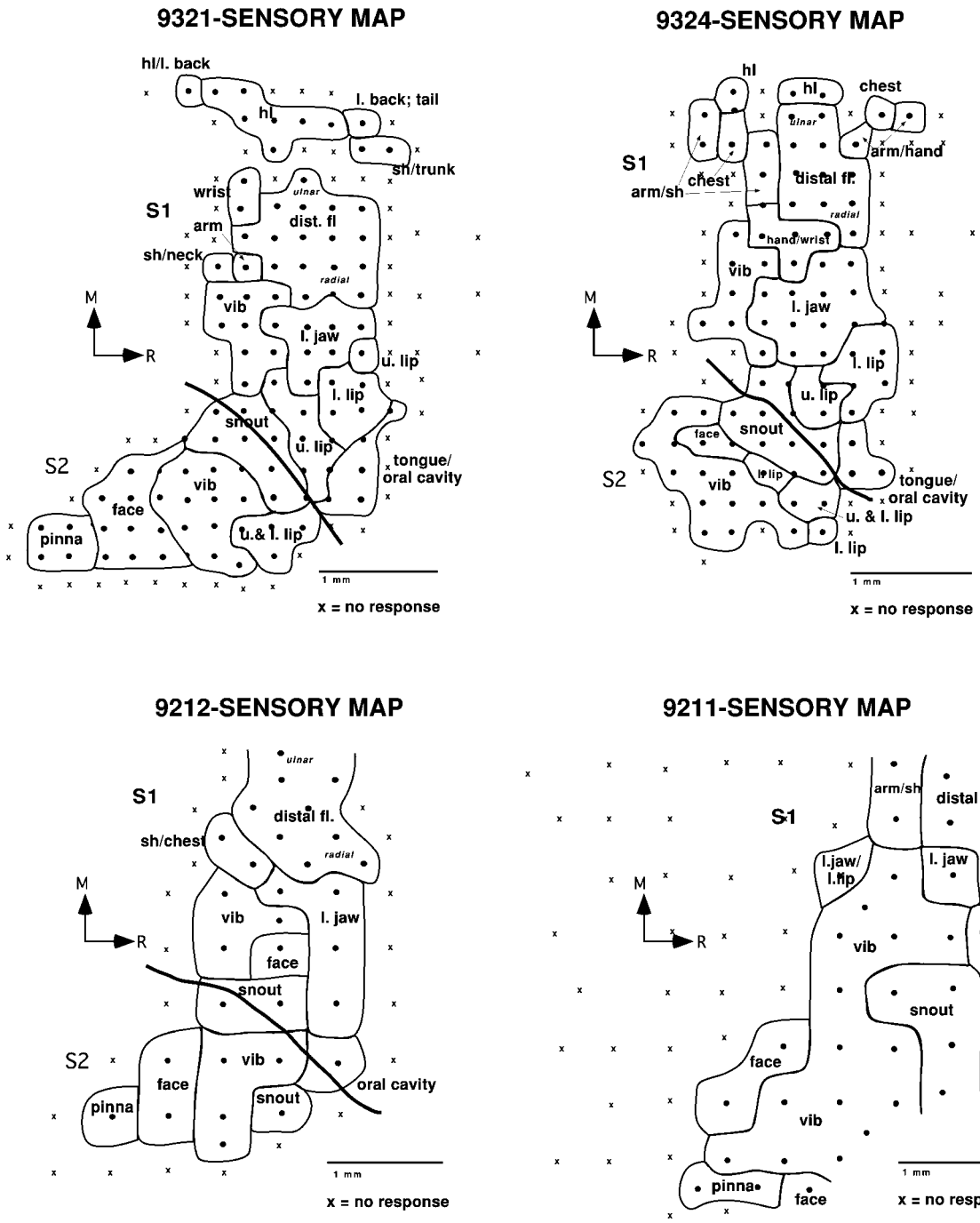


Fig. 2. Reconstructions of electrophysiological recordings showing the somatotopic organizations of primary (S1) and secondary somatosensory areas (S2) cortex in four *Monodelphis* opossums. The outer boundaries of S1 and S2 were estimated from the somatotopic patterns and receptive field sizes for each numbered case. Within each field, lines separate recording sites (dots) activated by a light touch on a designated body part. These results showed a common somatotopic organization of S1 with the hindlimb and trunk represented medially and the face and snout laterally. In opossums that demonstrated evidence of two somatotopic maps (9213; Fig. 1), 9321, 9324, and 9212) the transition from S1 to S2 was marked by a reversal of somatotopic order and an enlargement of receptive fields. This second somatotopic representation was positioned caudolateral to, and mirrored, the S1 representation. Somatotopic maps in each case occupied a similar rostrocaudal position in cortex (inset in Fig. 1).

Second somatosensory area, (S2). A second representation of the body surface was found to be situated caudolateral to S1. This second representation was apparent from the receptive fields of recording sites progressing caudolaterally from S1. The transition from S1 to S2 was typically marked by a reversal in body part representation

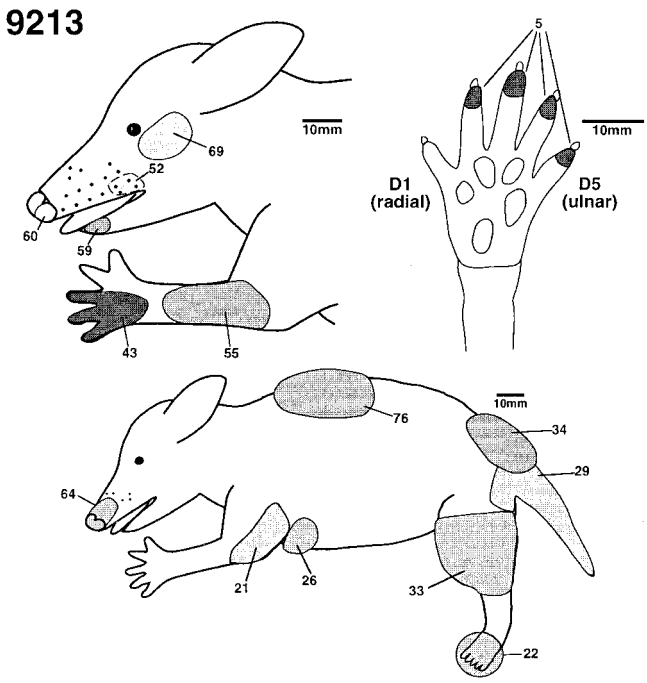


Fig. 3. Representative multi-unit receptive fields of somatosensory cortex in case 9213 (recording sites shown in Fig. 1). These are typical receptive fields found for the snout, face, vibrissae, digits, and forelimb and body. A large number of penetration sites were responsive to stimulation of the face and vibrissae, with most sites responsive to groups of vibrissae (site 52) and often entire rows. Receptive fields of the distal forelimb often included multiple digits and digit-hand combinations (sites 5 and 43). Penetrations that responded to stimulation of the back and lower body had receptive fields that were typically large in size (22, 33, 76, 29). Sites 64, 69, and 76 are located in secondary somatosensory area (S2), all others are in primary somatosensory area (S1).

and an increase in receptive field size (Fig. 5A–H). An enlargement of the receptive field size and a reversal of the somatotopic order are characteristics of neuronal responses in rows of recording sites crossing from S1 to S2 in several species (Krubitzer and Kaas, 1990; Beck et al., 1996). However, in the present study, receptive field sizes were occasionally similar in S1 and S2 and the border between fields was then determined by a somatotopic reversal. Receptive fields for progressively more lateral sites in S1 began in the posterior lower face, progressed to the mystacial vibrissae and ended on the glabrous snout (Fig. 5A–D). The receptive field at the next site (Fig. 5E) in the lateral progression reversed back to the mystacial vibrissae and was larger, including almost all the vibrissae on the contralateral face. Subsequent sites in the lateral progression had large receptive fields reversing back to the lower jaw vibrissae (Fig. 5G), to the face, head and neck (Fig. 5H,N) and to the entire hand (Fig. 5I) or shoulder and back (Fig. 5O).

This mediolateral transition from S1 to S2 is illustrated for case 9324 in Figure 6. Receptive fields for progressively more lateral sites in S1 began in the posterior mystacial vibrissae (Fig. 6, site 1), progressed to more anterior vibrissae (Fig. 6; 3, 4) and ended on the glabrous snout (Fig. 6; 5). The receptive field in the next site in the lateral progression was much larger and included the entire rostral snout and part of the upper lip (Fig. 6; 7). Still more laterally the receptive fields moved to the lower lip and jaw (Fig. 6; 8), followed by large region of the mystacial vibrissae (Fig. 6; 9). In a mediolateral progression along the rostral border of the somatosensory map receptive fields moved in an anterior direction along the lower lip to the ventral glabrous snout (Fig. 6A–C), then to the entire rostral snout and upper lip (Fig. 6D), followed by the entire upper and lower lip (Fig. 6E). In Figures 1 to 6, the delineation of the S1/S2 border is based on the relative size of the receptive fields at each site and the reversal of somatotopic representation in the caudolateral progression of recording sites.

Electrical stimulation maps of somatosensory cortex

Electrical stimulation maps of motor outputs from the physiologically identified somatosensory cortex were derived by using three different techniques as described in Materials and Methods: ICMS, low-impedance depth stimulation, and bipolar surface stimulation. The results using each of these techniques is described, in turn, below.

Intracortical microstimulation (ICMS). By using ICMS, sites throughout the entire cortical extent of the physiologically defined somatosensory cortex were explored for stimulus-evoked movements of skeletal musculature. In addition, sites immediately rostral and caudal to the somatosensory cortex were examined for stimulus-evoked responses. In most cases, movements were evoked by stimulation at sites within S1, but at no sites within S2. One exception was case 9211, where responses were evoked in more lateral portions of the sensory-response area that may be S2 (see Fig. 13). In most cases, ICMS evoked movements of vibrissae only, with an average threshold of 50 μA and ranging from a low of 9 μA to the maximum current level of 100 μA. Figure 7 shows the topographic location of ICMS sites in case 9213 and their location relative to previously identified corticospinal somata (Fig. 7, inset; Nudo and Masterton, 1990a). In this case, ICMS procedures resulted in movement of vibrissae at 13 sites and movement of the lower jaw at one site. In each of the other cases, ICMS procedures resulted only in movement of mystacial and lower jaw vibrissae. In the five cases in which complete, or nearly complete, ICMS-evoked movement maps were derived, the excitable area averaged 3.3 ± 2.1 mm^2.

Figure 8 shows the results of ICMS mapping in four additional animals. In cases where sensory recordings were also conducted, these areas largely overlapped areas responsive to cutaneous stimulation of the vibrissae, snout, face, and lower jaw (see input-output coupling below). The majority of these responses ($72.2 \pm 15\%$) were contralateral to the site of the stimulation, though ICMS also evoked ipsilateral ($16.6 \pm 7\%$) and bilateral ($11.4 \pm 12\%$) vibrissae movements. The contralateral, ipsilateral, and bilateral response sites appear to be clustered into distinct zones. In most cases, the contralateral responses were clustered in a caudomedial position relative to the more rostrolaterally placed ipsilateral response zone, and the more rostrally positioned bilateral response zone. ICMS typically evoked movement of all mystacial vibrissae or all lower jaw vibrissae on one side, though at many sites, movement of groups of mystacial vibrissae was observed. These were composed of two or more caudal, middle, or rostral vibrissae, or dorsal or ventral groups, and

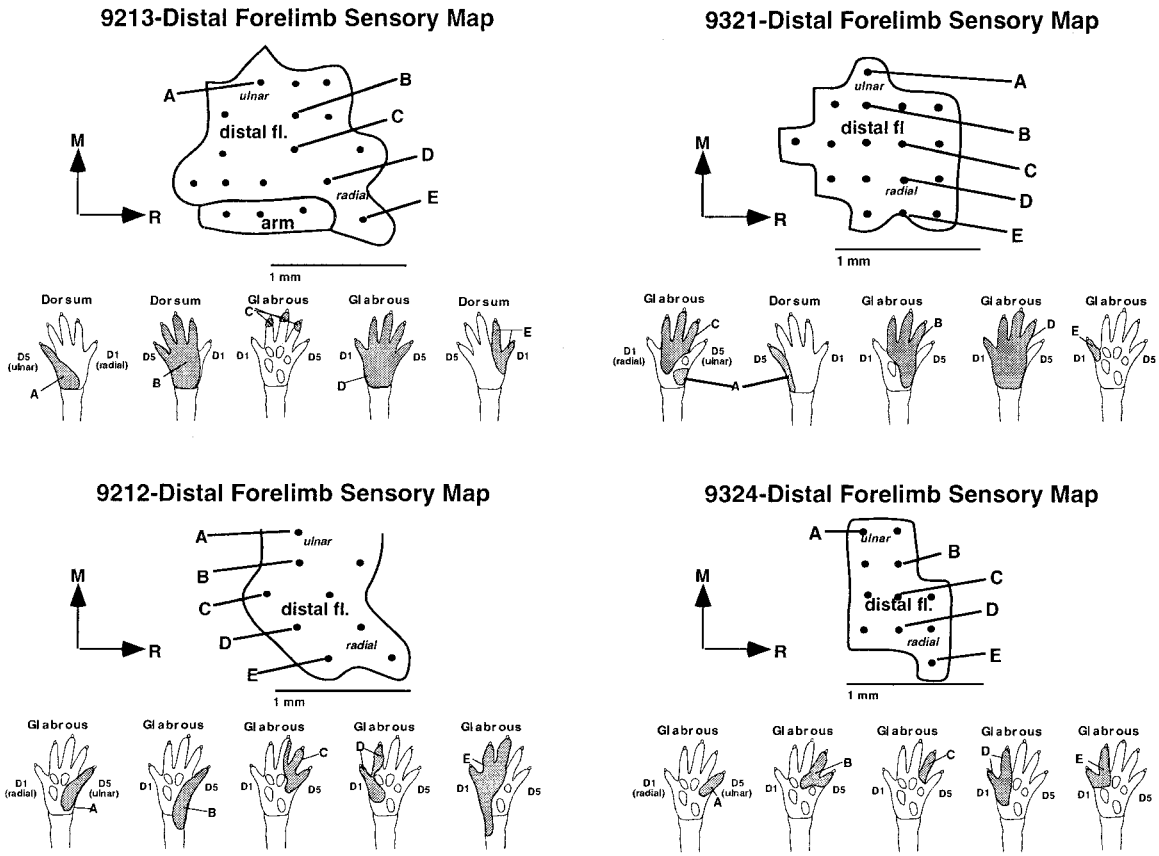


Fig. 4. The mediolateral topographic sensory representation of the distal forelimb in primary somatosensory area (S1) in four animals. The receptive fields are lettered according to the recording sites indicated on the surface view of the S1 distal forelimb area. In a line progressing from medial to lateral, receptive fields shifted from ulnar to radial aspects of the forelimb.

TABLE 1. Area Measurements (in mm^2) and Percentages of Body Part Representations Within Somatosensory Cortex[1]

Case no.	Tot. SS	S1	S2	Digits	Vibrissae	Rostrum	HL/trunk
9213	7.12	5.06	2.06	1.38	0.89	2.22	0.93
		(71%)	(29%)	(27%)	(18%)	(44%)	(18%)
9321	6.45	4.12	2.33	1.03	0.48	2.19	0.75
		(64%)	(36%)	(25%)	(12%)	(53%)	(18%)
9324	5.49	3.81	1.68	0.68	0.45	2.18	0.63
		(69%)	(31%)	(18%)	(12%)	(57%)	(17%)
Average	6.35 ± 0.82	4.33 ± 0.65	2.02 ± 0.33	1.03 ± 0.35	0.61 ± 0.25	2.20 ± 0.02	0.77 ± 0.15
		(68.0 ± 3.6%)	(32.0 ± 3.6%)	(23.3 ± 4.7%)	(14.0 ± 3.5%)	(51.3 ± 6.7%)	(17.7% ± 1.0%)

[1]The three cases shown had the highest resolution, the most definable boundaries and were considered fairly complete. Percentages of S1 and S2 areas are relative to Tot. SS. Area measurements of the digits, vibrissae, rostrum, and hindlimb/trunk are taken from the S1 area only and percentages reflect the proportion of that representation within S1. For abbreviations, see list. Table was produced by using Canvas 3.5.5.

occasionally various combinations of each (see Fig. 12 for examples). Although examined in detail, the evoked movements of vibrissae did not have a consistent topographic organization in any of these cases with respect to rostral/caudal or dorsal/ventral groups.

The motor representation was quite variable in size across animals, ranging in size from 1.21 to 6.25 mm^2. In addition, there was a tendency for males (n = 3) to possess a larger motor area than females (n = 2). Map areas in females averaged 1.53 ± 0.45 mm^2 and map areas in males averaged 4.47 ± 1.94 mm^2. However, this gender difference was not statistically significant, perhaps due to a small number of complete maps on which this data is based, and a large variance in the representational area, especially in males. Due to a substantial difference in size between females (45–67 g) and males (89–139 g), this gender difference also was examined after normalizing for body weight. Females had smaller map areas relative to males, even when corrected for body size, though this difference was again statistically nonsignificant (2.73 ± 0.5 mm^2/100 g in females; 4.64 ± 2.68 mm^2/100 g in males).

Low-impedance depth stimulation. In cases where low-impedance depth stimulation was employed, stimulation thresholds were greater than those found with ICMS procedures, ranging from a minimum of 90 μA to a max-

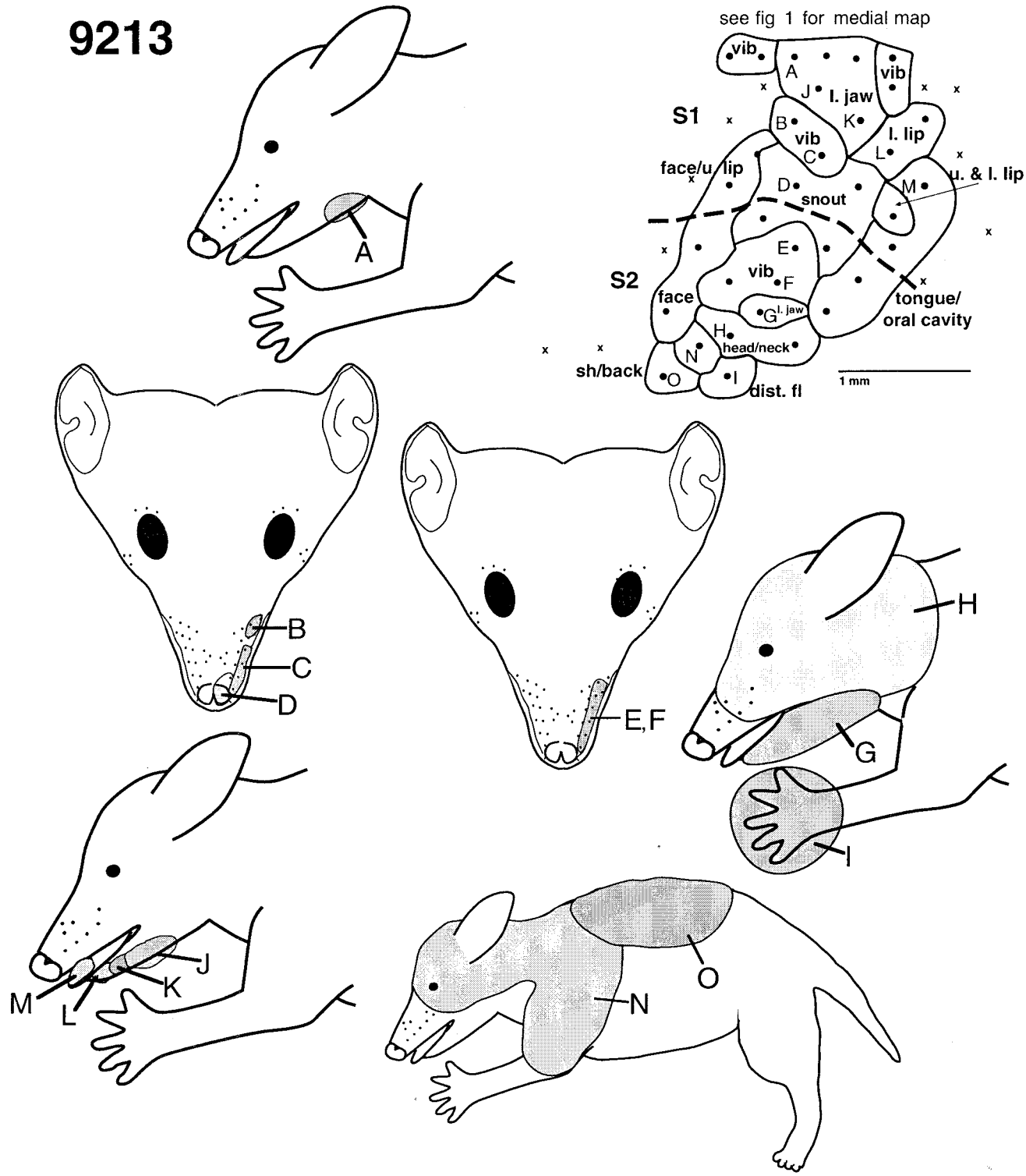


Fig. 5. Somatotopy of face representation and reversal of somatotopy across the primary/secondary somatosensory area (S1/S2) border. Receptive fields for two rows, one progressing across S1 (A and J–M), and one progressing across the border between S1 to S2 (A–I). A crude somatotopy was evident in a caudomedial to rostrolateral progression across S1, with receptive fields on the posterior aspect of the lower jaw located caudomedially, and receptive fields on the rostral aspect of the lower jaw, the snout, and oral cavity located rostromedially. In penetrations progressing from medial to lateral across the S1/S2 border (A–I), receptive fields moved from the lower jaw to the caudal mystacial vibrissae and to the snout, followed by a reversal back to the mystacial vibrissae, lower jaw, and head and neck, then to the distal forelimb. Receptive fields were generally larger in the S2 representation.

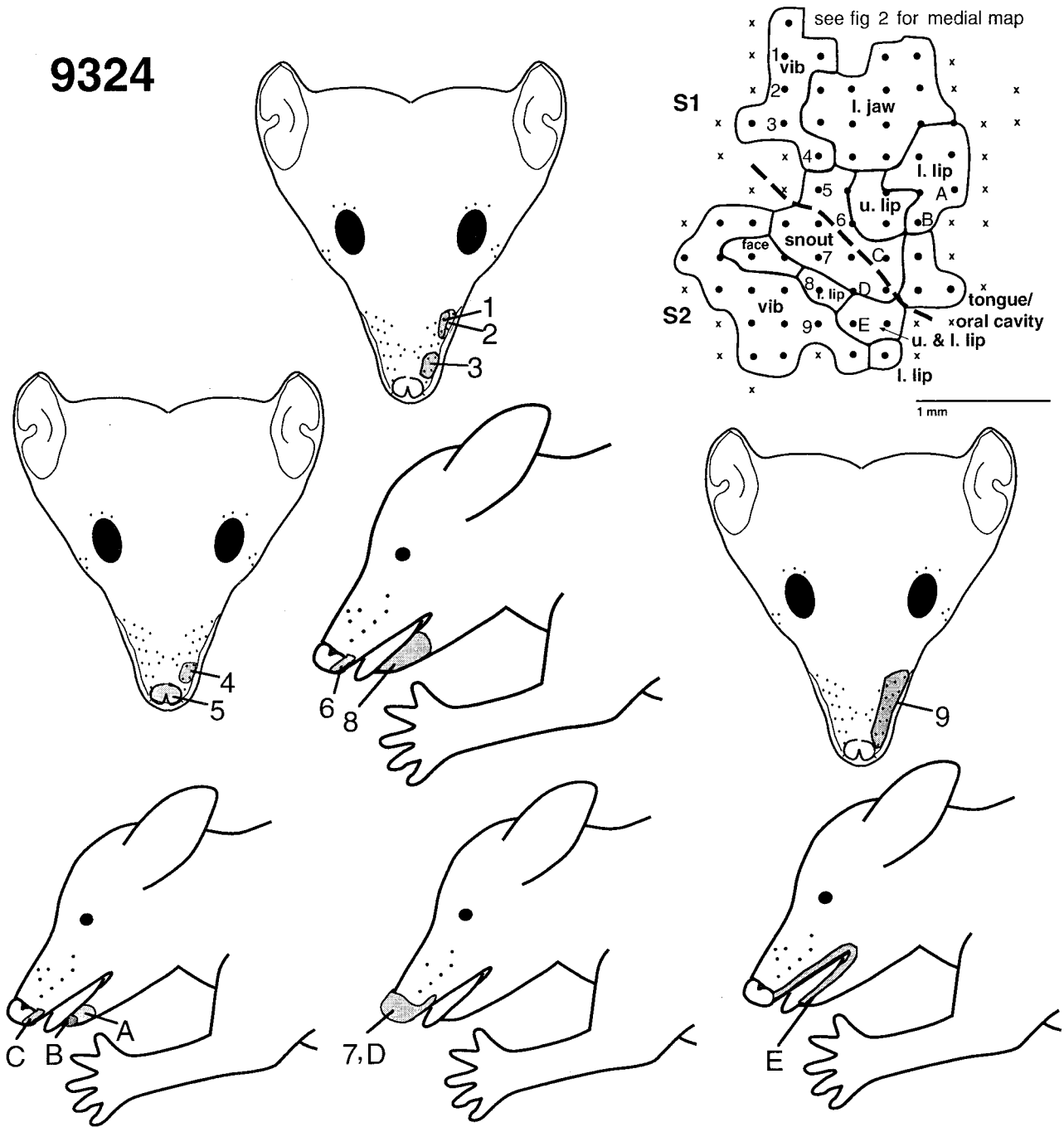


Fig. 6. Somatotopy of face representation and reversal across the primary/secondary somatosensory area (S1/S2) border. Receptive fields for neurons in electrode penetrations in two rows (1–9) and A–E. In penetrations progressing in a caudomedial to rostrolateral direction within S1, receptive fields moved from the posterior mystacial vibrissae (1) to the glabrous snout (5, 6). Continuing across the S1/S2 border, receptive fields then increased in size to include the entire rostral snout and part of the upper lip (7) followed by a return to the mystacial vibrissae (9). In a second line along the rostral portion of S1, receptive fields of sites in a medial to lateral progression moved from the lower lip to the glabrous snout (A–C). Then, progressing across the S1/S2 border a large representation of the rostral snout (D) was encountered and then the entire upper and lower lips (E). The crude somatotopic reversal and increase in receptive field size was used to delineate the S1/S2 border.

imum of 300 μA. Only vibrissae movements were evoked using this procedure. The majority of these responses (70.3%) were contralateral to the site of stimulation, similar to that seen using ICMS, although in one case (9321, Fig. 9) most responses (89%) were ipsilateral. As defined by low-impedance depth stimulation, the vibrissae motor

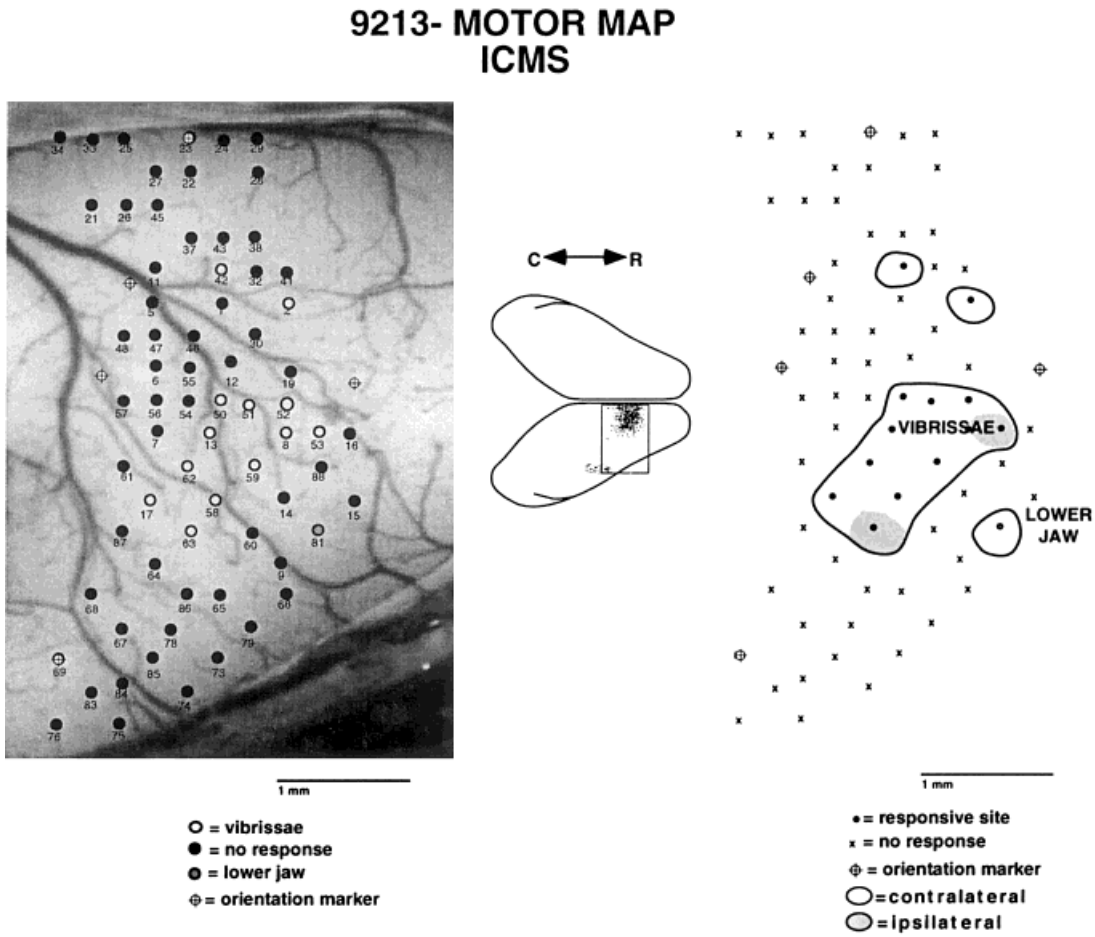


Fig. 7. Scope of intracortical multi-unit stimulating (ICMS) experiments in case 9213. Mapping procedures were guided by the locations of corticospinal neurons (**inset**) derived in previous experiments (Nudo and Masterton, 1990). Stimulation of cortex resulted in movement of vibrissae and in a single penetration, movement of the lower jaw. These sites overlapped sensory sites that were responsive to cutaneous stimulation of the body surface on the face, snout and lower jaw, and to a lesser extent stimulation in the digit area. The majority of responses were contralateral to the site of stimulation (unshaded area). Two rostral penetrations evoking movement of ipsilateral vibrissae (shaded area).

representation typically overlapped with sensory representations of the vibrissae, jaw, face, snout, and lips, similar to that seen in ICMS experiments, but also often extended well into the S2 sensory map (Fig. 9). In some cases, the motor representation also extended medially into the digit sensory area (not shown). The excitable cortical zone was slightly more extensive than the zone defined by ICMS (4.1 mm^2 in the most complete case, a female, 9321; Fig. 9), or 4.9 mm^2 per 100 g body weight). Incomplete maps in males disallowed for gender comparisons of areal size of the motor representation in the low-impedance maps.

Unlike ICMS, this procedure typically evoked movement only of small groups of mystacial vibrissae on one side and not the entire mystacial or lower jaw vibrissae field. These responses were composed of two or more adjacent caudal, middle, or rostral vibrissae, usually in one or two rows, and rarely combinations of each. No obvious topography of vibrissae motor representation was found in any of these low-impedance depth stimulation procedures with respect to rostral/caudal or dorsal/ventral groups.

Bipolar surface stimulation. In cases where bipolar surface stimulation was employed, stimulation thresholds were greater than that found with ICMS or low-impedance depth stimulation procedures, and ranged from a minimum of 0.6 mA to a maximum of 3 mA. Again, only vibrissae movements were evoked by using this procedure and response areas were highly variable across animals (Fig. 10). The majority of these responses (98 $\pm$ 1.5%) were contralateral to the site of stimulation, similar to that seen by using ICMS, with two cases having single bilateral response sites in each map. The excitable cortical zone using surface stimulation was considerably more extensive than the zone responsive to ICMS or low-impedance depth stimulation and averaged 9.1 $\pm$ 2.5 mm^2 in area, or 10.9 $\pm$ 6.2 mm^2 per 100 g body weight (Fig. 10). Although complete sensory and surface stimulation maps were not conducted in the same animals, the excitable area extended over what was observed to be most of S1 and a large portion of S2 in other cases. Vibrissae movements were again evoked at sensory vibrissae, face, jaw, lip, and oral cavity sites, and also some at distal forelimb

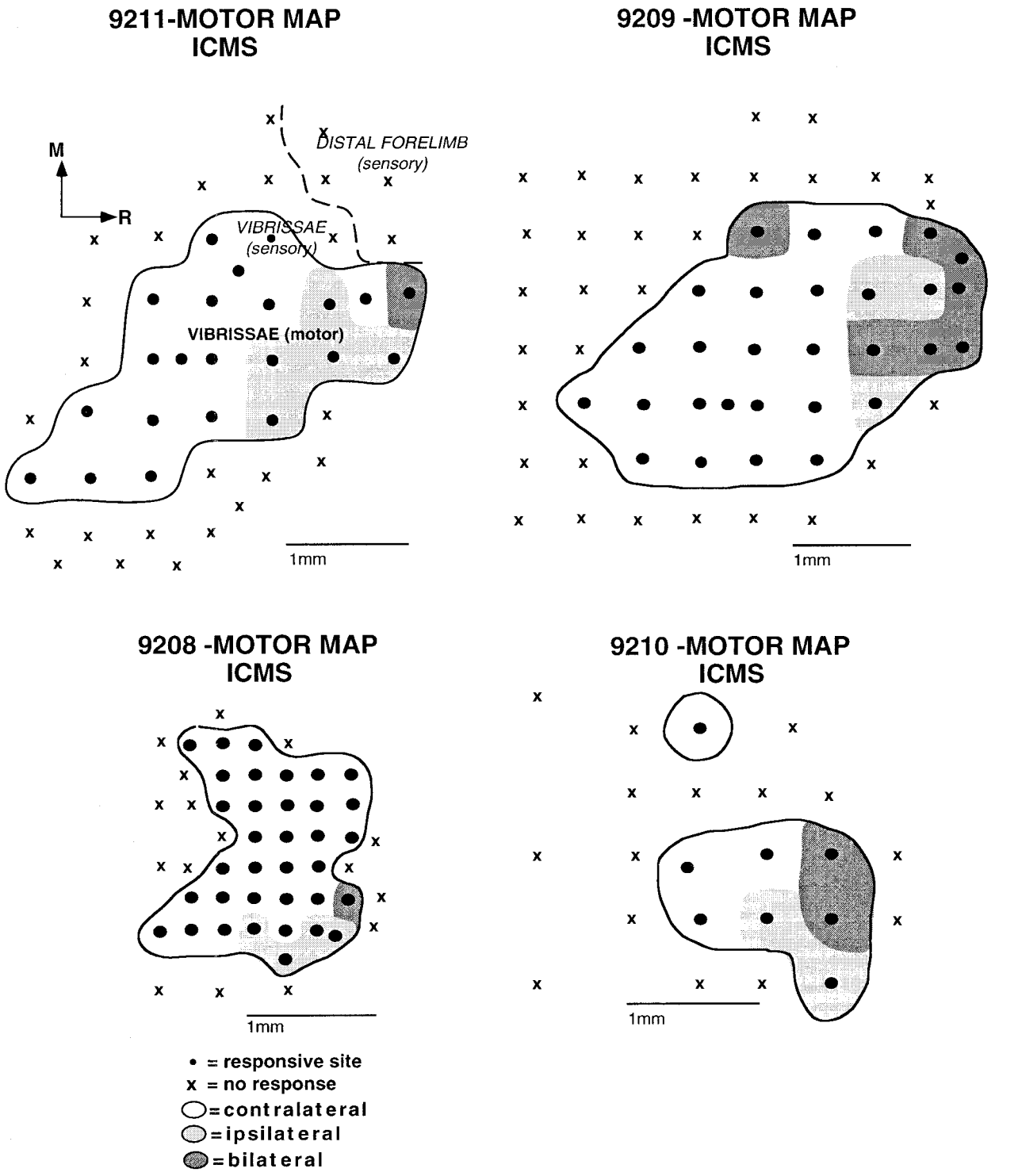


Fig. 8. Results of intracortical multi-unit stimulation in four *Monodelphis* opossums. In all four animals, only vibrissae movements were evoked following stimulation of cortex. In cases where sensory recordings were also conducted (9211, 9213; Fig. 7) these areas largely overlapped areas responsive to cutaneous stimulation of the vibrissae, snout, face, and lower jaw. Contralateral (no shading), ipsilateral (moderate shading), and bilateral (heavy shading) areas are clustered into distinct zones. Contralateral responses have the largest area and are clustered in a caudomedial position relative to the more rostrolaterally positioned ipsilateral zone, and the more rostrally positioned bilateral response zone.

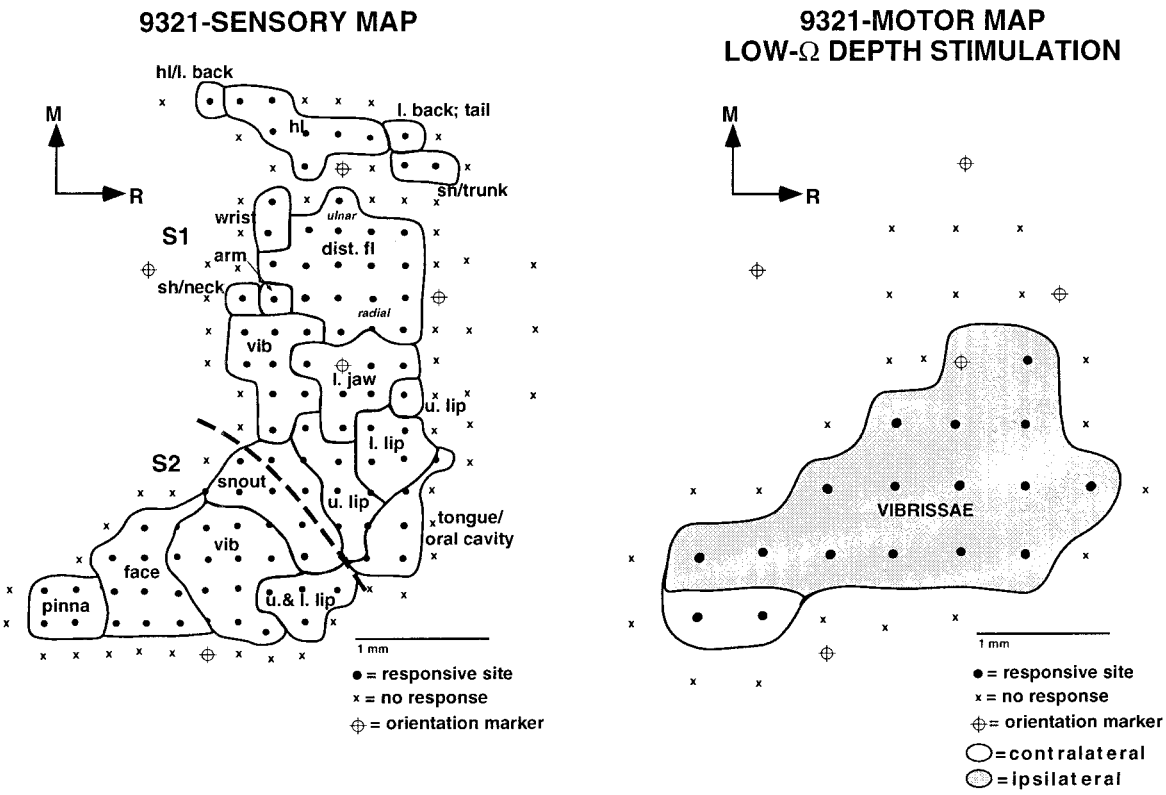


Fig. 9. Motor responses of cortex in animal 9321 evoked by using low-impedance depth stimulation (right) compared to the sensory responses in the same animal (left). Motor responses were confined to vibrissae movement. These sites overlapped the vibrissae, face, and jaw sensory areas (left). Overall areas that elicited motor responses following low-impedance depth stimulation were generally larger than seen with intracortical multi-unit stimulation (ICMS).

and even hindlimb sites. Similar to that seen with low-impedance depth stimulation, this procedure typically evoked movement only of small groups of mystacial vibrissae on one side and not the entire mystacial or lower jaw vibrissae field. These responses were composed of two or more adjacent caudal, middle, or rostral vibrissae, usually in one or two rows, and rarely combinations of each. No obvious topography or vibrissae motor representation was found in any of these surface stimulation procedures with respect to rostral/caudal or dorsal/ventral groups.

Input/output coupling. By using ICMS, it was possible to make specific comparisons between the sensory receptive fields and the evoked movements at each site. Results from two cases in which detailed ICMS and sensory mapping was conducted are shown in Figures 11 through 14. Figure 11 shows the results of ICMS mapping in case 9213 in relation to the sensory map derived in the same animal. Electrical stimulation of sites that evoked movement of vibrissae (shaded area) generally overlapped sensory sites at which neurons were responsive to cutaneous stimulation of the vibrissae, face, snout, and lower jaw, and to a lesser extent, stimulation of the distal forelimb. One site evoked movement of the lower jaw and was responsive to cutaneous stimulation of the tongue and oral cavity in sensory mapping. Electrical stimulation occasionally resulted in the movement of the same vibrissae whose deflection evoked neuronal activity at the same site in sensory mapping. However, in the majority of vibrissae motor sites (11 of 14), this was not the case. Furthermore, the degree of sensory/motor identity did not appear to be related to the current required for evoking the motor responses. For example, in case 9213 (Fig. 12), sites that had the lowest stimulation thresholds (9 and 13 μA) for evoked movements had matching vibrissae in sensory recordings, but a similar match was also found at a site that had a relatively high motor response threshold (67 μA).

Although not extensively explored, electrical stimulation of sites rostral to the somatosensory representation rarely resulted in evoked movements. In one case (9213; Fig. 11), a single site rostral to the sensory area was responsive to electrical stimulation with a relatively high threshold of 72 μA. In other cases in which both sensory and ICMS mapping was conducted, penetrations 250 μm rostral to the sensory map were routinely stimulated and no site rostral to the sensory area evoked movement upon stimulation.

In some cases, motor responses were also evoked at sites caudal to the physiologically defined somatosensory area. For example, in case 9211 the ICMS map again overlapped the sensory representation of the vibrissae, face, snout, and lower jaw, but in addition, vibrissae movements were evoked at several sites that were caudal to the somatosensory representation (Fig. 13). No obvious topography of vibrissae motor representation was found in case 9211, nor was there a trend toward direct coupling of particular vibrissae motor and sensory representation.

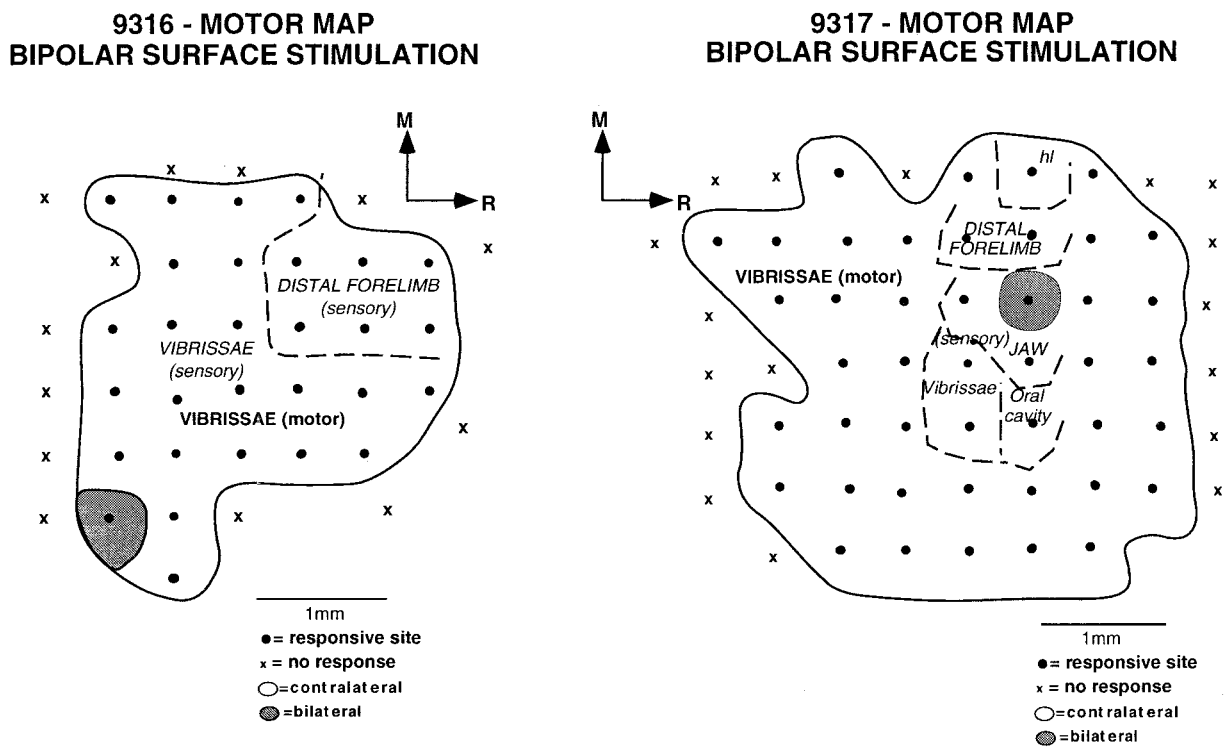


Fig. 10. Motor responses of cortex in two animals evoked by using bipolar surface stimulation. Stimulating sites are the midpoint between the two ball electrodes. Motor responses were again confined to vibrissae movement in both cases. In case 9316 (left), responses overlapped sensory representations of the vibrissae and distal forelimb when compared to low resolution sensory mapping (italics within dashed lines). In case 9317 (right), motor responses overlapped sensory representation of vibrissae, jaw, oral cavity, distal forelimb and hindlimb. Overall, motor maps were significantly greater in area by using bipolar surface stimulation as compared to typical results obtained by using intracortical multi-unit stimulation (ICMS), which usually did not evoke movement from distal forelimb sensory areas.

Only one site demonstrated both sensory and motor representation of the same vibrissae group, with a motor threshold of 45 μA (Fig. 14).

Cytoarchitecture

Nissl stains of coronal sections through somatosensory and motor-responsive cortex revealed a six-layered cortex with a discernible granule cell-dense layer 4 and a moderately packed pyramidal cell layer 5 (Fig. 15). Areas that were responsive to cutaneous stimulation and also evoked movement upon intracortical microstimulation did not appear to be cytoarchitectonically different in Nissl stains from areas that were exclusively responsive to sensory stimulation.

In tangentially sectioned cortex, the somatosensory cortex stained darkly for myelin (Fig. 16). Staining was similar in density in S1 and S2 and no discernible separation between these areas was apparent. A very small and light septa separated S2 from the auditory area that was located just above the rhinal fissure (Frost and Masterton, 1992; Melzer et al., 1997). Auditory cortex was the most darkly stained area and visual cortex located in the caudolateral quadrant was only lightly stained. These results are similar to that seen in *Didelphis* (Beck et al., 1996).

DISCUSSION

The present results suggest that somatosensory cortex in the *Monodelphis* opossum consists of at least two somatotopically organized fields, S1 and S2, similar in position and organization to those seen in other mammals. In contrast to results from other mammals, electrical stimulation mapping did not reveal a complete motor representation within somatosensory cortex. Virtually all evoked movements were movements of mystacial or lower jaw vibrissae. ICMS maps suggest that these evoked responses are predominantly confined to S1.

The primary somatosensory area (S1)

The position of S1 in *Monodelphis* is similar to that observed in *Didelphis* and other mammalian species (Lende, 1963b; Beck et al., 1996; see Kaas, 1983; Johnson, 1990). Also, the general somatotopic organization of S1 is similar to that shown in other mammals, such as *Didelphis* (Lende, 1963b; Pubols et al., 1976; Beck et al., 1996) rat (Woolsey, 1958; Hall and Lindholm, 1974), shrews (Catania et al., 1999), cat (Woolsey, 1958; Dykes et al., 1980; Felleman et al., 1981), insectivores (Catania and Kaas, 1995; Krubitzer et al., 1997; Pobirsky et al., 1998), tree shrew (Sur et al., 1980b), prosimians (Krishnamurti et al., 1976; Sur et al., 1980a), and primates (Merzenich et al., 1978; Sur et al., 1982; Krubitzer and Kaas, 1990; see Kaas, 1983 for review). The existence of S1 has also been documented in several Australian marsupials (Johnson et al., 1973; Johnson, 1990; Elston et al., 1993; Weller, 1993; Huffman et al., 1999), with a somatotopic organization similar to that found in the Didelphid opossums. As in most other mammals, S1 in *Monodelphis* contains a com-

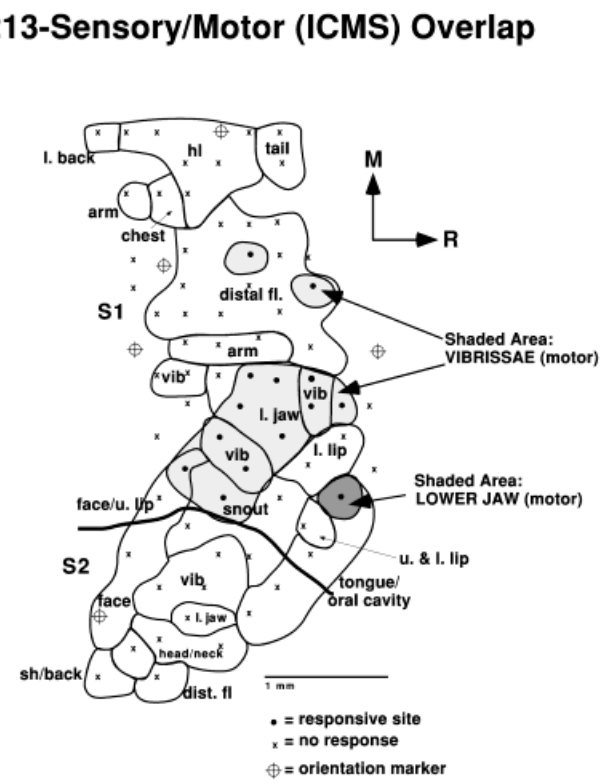


Fig. 11. Results of intracortical multi-unit stimulation (ICMS) in case 9213 in relation to the sensory map in the same animal. Stimulating sites that evoked movement of vibrissae generally overlapped sites that were responsive to stimulation of the vibrissae, lower jaw, face, and snout. Two sites that, when stimulated evoked vibrissae movement, were responsive to stimulation of the distal forelimb. One site evoked movement of the lower jaw and was responsive to stimulation the tongue and oral cavity.

plete representation of the contralateral body surface, including the hindlimb and tail, the trunk, the forelimb, and the face in a mediolateral sequence. The somatotopic organization of the digit representations is also similar to that in other mammals, with the ulnar digits represented most medially and the radial digits represented most laterally (see Kaas, 1983 for review). The results reported here are similar to those of a recent study with less detailed mapping that found a similarly organized S1 in *Monodelphis,* but no evidence of an S2 representation (Huffman et al., 1999).

The very large rostrum representation within S1 is also seen in *Didelphis* (Lende, 1963b; Pubols et al., 1976; Beck et al., 1996). This large representation of the rostrum may reflect its behavioral importance in Didelphids, just as the large representation of the bill in the duck-billed platypus (Krubitzer et al., 1995) and large snout representation in the star-nosed mole (Catania and Kaas, 1995) reflect specialized usages.

The secondary somatosensory area (S2)

The present results provide evidence for a second somatosensory representation of the body, positioned lateral to S1. This body representation is oriented similarly to S2 reported in *Didelphis* (Lende, 1963b; Pubols, 1977; Beck et al., 1996), as well as other mammalian species, with a mirror reversal of the organization found in S1 (Krubitzer et al., 1986; Johnson, 1990; Rowe, 1990; Krubitzer, 1996). A secondary somatosensory area has been identified in several Australian marsupials, including the brush-tailed possum (Elston et al., 1993; Coleman et al., 1999) and the striped opossum (Huffman et al., 1999). In addition to S1, at least one lateral somatosensory field corresponding to S2 has been identified in virtually all mammals studied, suggesting that these fields were retained from a common ancestor and fit a basic mammalian plan. In *Didelphis,* a third representation responsive to cutaneous stimulation is found lateral to S1 and rostroventral to S2, and is believed to be homologous to the parietal ventral area (PV) in eutherian mammals (Beck et al., 1996; see Krubitzer et al., 1993 for review). Two additional representations responsive to stimulation of deep receptors are positioned immediately rostral and caudal to S1, and referred to as the rostral and caudal somatosensory areas, SR and SC. The present results did not reveal the existence of these additional representations, although they were sought. The second representation shown here is believed to be a homologue of S2 in *Didelphis* because of its orientation with the distal forelimb lateral and mouth rostral (Fig. 1), whereas in the PV representation in *Didelphis* and other mammals the distal forelimb is medial and the mouth is caudomedial to the body representation (Krubitzer et al., 1993; Beck et al., 1996). If the *Monodelphis* organization is accepted as the more primitive condition from which *Didelphis* organization is derived (see Introduction), evidence of only two somatosensory fields in *Monodelphis* and a total of five in *Didelphis* may support the long-held belief that multiple fields represent the differentiation of new cortical areas from preexisting areas (Brodmann, 1909; von Economo, 1929; Kaas, 1987, 1989).

A PV representation and two additional rostral fields responsive to stimulation of deep receptors are seen in present-day monotremes, the short-billed echidna, and platypus (Krubitzer et al., 1995). The presence of a PV representation in monotremes that diverged from therian mammals over 130 million years ago (before the appearance of Didelphids; Clemens, 1989; Flannery, 1989; Westerman and Edwards, 1992) and apparent lack of a PV representation in *Monodelphis* suggests that PV (as well as additional deep receptor fields) may have evolved independently in monotremes and *Didelphis.* Alternatively, it is possible that PV is vestigial in *Monodelphis.* The present results cannot adequately resolve this issue, because the spatial resolution of the present maps was relatively low in the most ventral aspects of the explored region. Higher resolution maps in this region may yet reveal a third somatosensory area in this species.

Motor representation

The present results show that only vibrissae and in one case lower jaw movements were evoked upon electrical stimulation of S1 in *Monodelphis.* This result was the same regardless of the stimulation technique used. ICMS, low-impedance depth stimulation, and bipolar surface stimulation produced no movements caudal to the lower jaw. Results using the latter technique are especially revealing, because they were identical to those used by Lende to show a complete sensorimotor amalgam in *Didelphis* opossums (Lende, 1963a,c). These relatively more crude techniques result in stimulation of a much greater

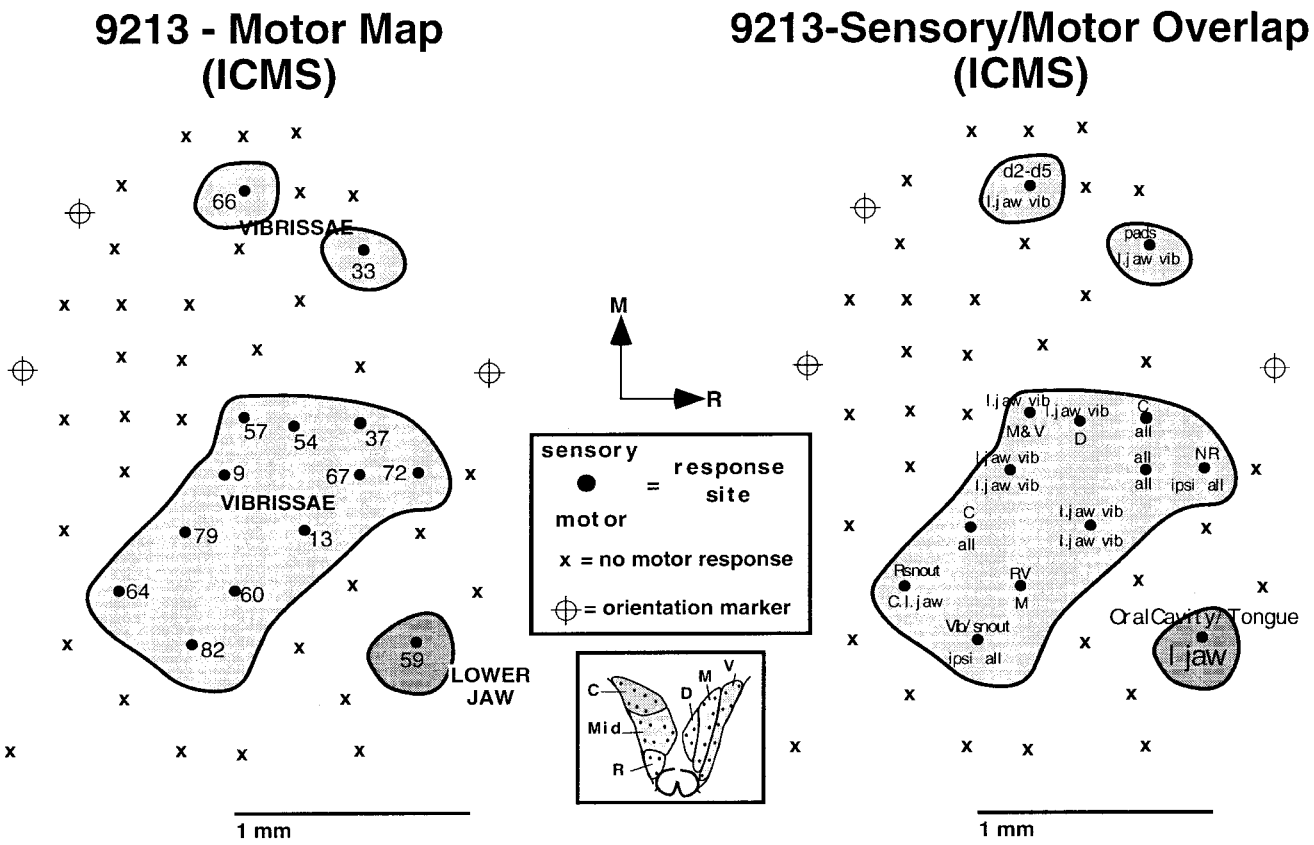


Fig. 12. Input/output coupling: matching vibrissae motor representation to sensory representation in animal 9213. Left, numbers next to the stimulating sites that evoked movement indicate the threshold for that movement (in μA). Right, motor responses are abbreviated below the stimulating site; sensory receptor fields are abbreviated above the recording site in the same penetration. Most vibrissae motor response areas overlapped sensory areas responsive to vibrissae, snout, face, and lower jaw vibrissae. Several sites that, when stimulated evoked movements of vibrissae, were also responsive to stimulation of the same vibrissae. The sites with the lowest intracortical multi-unit stimulation (ICMS) response threshold (9 and 13 μA) were sites that were both responsive to lower jaw vibrissae deflection and when stimulated could evoke movement of the same vibrissae. One site with a high threshold (67 μA) also had similar vibrissae sensory and motor representation. RV, rostroventral. All responses are contralateral unless indicated.

volume of tissue and provide synchronous activation of a much larger number of corticofugal neurons, maximizing the effects of a sparse projection pathway to the spinal cord.

In eutherian species, M1 is situated rostral to S1 and has a parallel somatotopic organization (e.g., Gould et al., 1986). No motor representation of body parts below the level of the head was found in S1 or in more rostral areas in the present study. This is in sharp contrast to the results reported by Lende (1963a) in *Didelphis,* where a complete and coincident overlap of somatic motor and somatic sensory representations was found. Results using similar techniques have shown a motor representation superimposed on S1 in *Didelphis azarae* (Magalhães-Castro and Saraiva, 1972). In both studies, the amalgam was quite large, encompassing the complete rostrocaudal length of the parietal region, perhaps due to the use of surface recording and stimulation techniques. Motor maps using bipolar surface stimulation reported here were also quite large, an average of 2.75 times larger than ICMS maps (see Fig. 10). More recent examination of motor responses in *Didelphis* cortex by using ICMS found high-threshold (up to 100 μA) response sites within, and just rostral and caudal, to S1 (Beck et al., 1996). Stimulation of cortex by Beck et al. (1996) most often resulted in movement of the tongue, though evoked movements of body parts, including several forelimb movements, were also seen.

That electrical stimulation of a large area of neocortex in *Monodelphis* evokes movement only of vibrissae suggests that evolutionary pressures for development of motor components in cerebral cortex began with a need for more refined movements of the face. This pressure may have resulted in an early substrate for mediating vibrissae movement, long before sufficient corticospinal pathways were in place to mediate refined control of the forelimbs and hindlimbs, and long before a separate motor representation rostral to S1 was established. Similar to the large somatosensory representation of the rostrum, this large vibrissae motor representation may be related to the behavioral importance of the area in *Monodelphis.* Behavioral studies have shown that *Monodelphis* exhibit a behavior wherein the rostral portion of the face is used for scent marking (Fadem and Cole, 1985; Fadem and Corbett, 1993) and the anteroventral portion of the glabrous snout is used to deliver odors to the vomeronasal

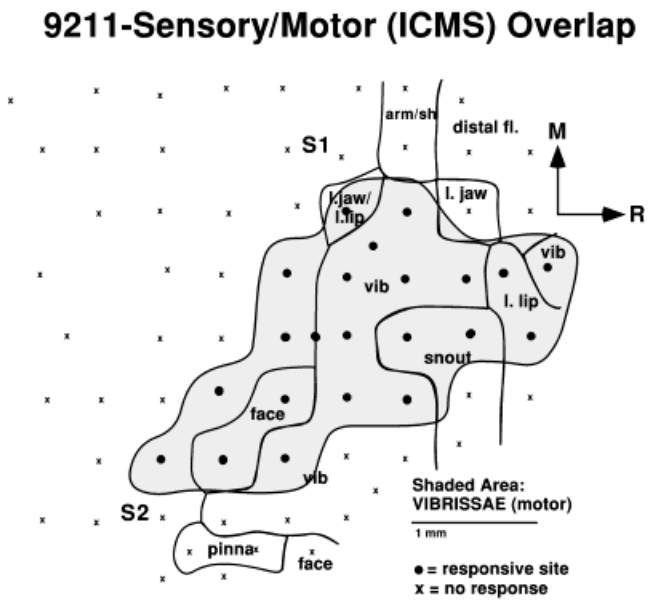


Fig. 13. Results of intracortical multi-unit stimulation (ICMS) of animal 9211 in relation to the sensory map in the same animal. Stimulating sites that evoked movement of vibrissae generally overlapped sites that were responsive to stimulation of the vibrissae, lower jaw, face, and snout. Several motor response sites were found caudal to the sensory representation.

organ (Poran et al., 1993a,b). This latter "nuzzling" behavior is thought to be similar to pheromone delivery behaviors seen in several placental and marsupial species and may represent a primitive mammalian pattern that has been retained in some species over phylogeny (Kimble, 1997). Furthermore, scent marking utilizing the face, is more commonly exhibited in males (Fadem and Cole, 1985; Fadem and Corbett, 1993). This gender-related behavioral difference may be related to the possible larger size of the vibrissae motor response area found in male *Monodelphis* as compared to female *Monodelphis* in the present study.

It has been suggested that vibrissae are among the phylogenetically youngest of the specialized mammalian peripheral sensory organs, lacking in monotremes and emerging in theria about 120 million years ago (Pocock, 1914; Brecht et al., 1997). Although most studies of the vibrissae have examined their sensory roles in behavior, it is clear that mystacial vibrissae are active during exploratory and discriminative behaviors, at least in rats (Welker, 1964; Carvell and Simons, 1990, 1995; Bermejo et al., 1996; Kleinfeld et al., 1999). Results from behavioral studies in the rat suggest that the large mystacial vibrissae (macrovibrissae) are involved in spatial perception (Hutson and Masterton, 1986) and act as distance decoders (Brecht et al., 1997). Any advantage that refined movement of the vibrissae could provide to an individual would have contributed to the evolutionary pressures for their cortical motor representation. Furthermore, it has been suggested that the smaller vibrissae (microvibrissae) on the rostroventral portion of the snout constitute an object-recognition organ, possessing a large receptor density crucial for tactile object recognition (Brecht et al., 1997). This view fits well with the findings that the rostroventral snout is important in nuzzling behavior (Poran et al., 1993a).

It has long been established that movement can be evoked from electrical stimulation of S1 in primates (Schafer, 1900; Penfield and Boldrey, 1937; Penfield and Rasmussen, 1952; Woolsey et al., 1953; Woolsey, 1958) and cats (Woolsey, 1947). Similarly, Malis et al. (1953) found that sensory-evoked potentials could be elicited in precentral motor cortex in monkeys. More recent electrophysiological studies have confirmed somatic motor functions in S1 (Doetsch and Gardner, 1972; Sapienza et al., 1981; Gioanni and Lamarche, 1985) and somatic sensory functions in M1 (Welt et al., 1967; Thompson et al., 1970; Rosén and Asanuma, 1972; Wiesendanger, 1973; Asanuma et al., 1979, 1980; Tanji and Wise, 1981; Strick and Preston, 1982; Asanuma and Arissian, 1984; Nudo et al., 1997).

In general, electrically evoked motor responses in S1 resemble those evoked in M1, with the exception that stimulation thresholds are greater (Welker et al., 1957; Woolsey, 1958; Doetsch and Gardner, 1972; Sessle and Weisendanger, 1982). In both the results presented here and results reported in electrical stimulation studies of cortex in *Didelphis* (Lende, 1963c; Beck et al., 1996), movements were most often evoked by using relatively high threshold stimulation; higher than the < 20 μA typically observed in microstimulation studies of M1 of primates (Sessle and Weisendanger, 1982; Stepniewska et al., 1993; Nudo et al., 1992, 1996). This disparity in thresholds for evoked movements suggests that the electrically evoked motor responses observed in *Monodelphis* are representative of the motor component of somatosensory cortex found in other mammals.

Laterality of responses. ICMS and unit recordings in a macaque monkey have shown that ipsilateral and bilateral movements of the hand are represented anterolateral to regions associated with the more common contralateral movements (Aizawa et al., 1990). Functional magnetic resonance imaging in humans has shown that segregated ipsilateral response areas are anterolateral to contralateral response areas (Cramer et al., 1999). That ICMS ipsilateral and bilateral vibrissae zones are segregated similarly in *Monodelphis* may suggest that this organization developed early in the appearance of motor-responsive areas of cortex and has been retained through the evolution of primate motor cortex.

It is unclear why a consistent segregation of response zones was not observed in low-impedance or depth stimulation mapping. The most complete low-impedance stimulation case had predominantly ipsilateral vibrissae responses (Fig. 9) and may have been an anomalous animal.

Input-output relationships. The finding that vibrissae motor response sites typically overlapped sites with sensory receptive fields of the vibrissae and other parts of the face is roughly similar to that found in somatomotor cortex of other mammals. Cortical sites receiving tactile input from particular parts of the face, hand, or trunk in monkeys roughly overlap sites that result in movements in closely related parts of the body upon stimulation (Woolsey, 1958; Doetsch and Gardner, 1972). A similar rough congruence of sensory input and motor output is seen in sensorimotor cortex of rats (Sapienza et al., 1981) cats (Asanuma et al., 1968), and primates (Rosén and Asanuma, 1972; Sessle and Weisendanger, 1982; Strick and Preston, 1982).

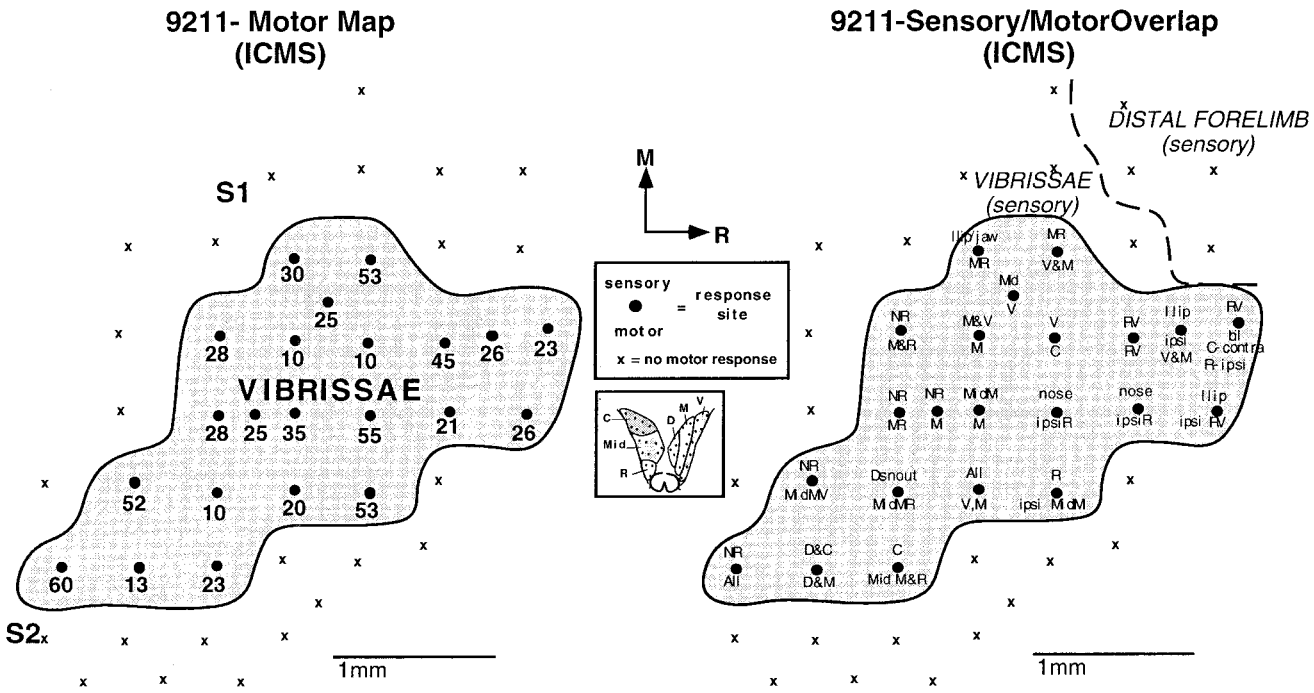


Fig. 14. Input/output coupling: matching vibrissae motor representation to sensory representation in animal 9211. Left, numbers indicate motor response thresholds in μA. Right, motor responses are abbreviated below the stimulating site; sensory receptive fields are abbreviated above the recording site in the same penetration. All intracortical multi-unit stimulation (ICMS) driven responses were vibrissae movement. Motor responsive sites generally overlapped sensory representation of the vibrissae, lower jaw, face, and snout. A direct coupling of particular vibrissae motor and sensory representation was seen at only one site, with a motor threshold of 45 μA. D, dorsal; V, ventral; M, vibrissae medial to the ventral rows and lateral to the dorsal group; C, caudal group; R, rostral group; Mid, vibrissae group between caudal and ventral groups. All responses are contralateral unless indicated.

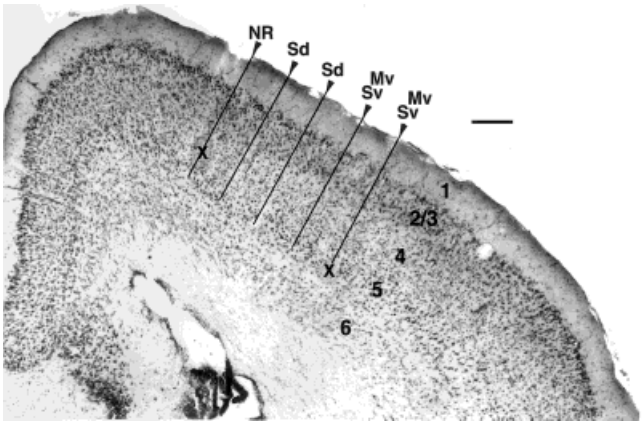


Fig. 15. Coronal section through the forebrain of *Monodelphis* 9319 counterstained with cresyl violet. Arrowheads indicate locations of microelectrode penetrations at the forelimb/vibrissae sensory border. X, electrolytic marking lesion; Sd, sensory digit response; Sv, sensory vibrissae response; Mv, motor vibrissae response; NR, no response. Numbers indicate cortical layer. Figures 15 and 16 are in Adobe Photoshop 4.0. Scale bar = 250 μm.

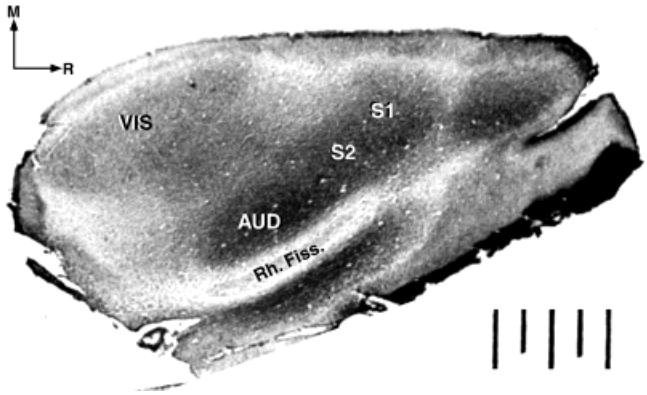


Fig. 16. Lightfield photomicrograph of *Monodelphis* cortex that has been flattened, sectioned tangentially, and stained for myelin. The somatosensory area can be distinguished as a myelin dark region. Distance between scale bars = 1 mm.

Contributions of the corticospinal tract to control of dexterous forelimb movement

Monodelphis possesses only about 500 corticospinal neurons (Nudo et al., 1995). What few corticospinal neurons *Monodelphis* does have, terminate in the cervical and perhaps the thoracic spinal cord, and do not appear to reach the lumbar or sacral cord (Holst et al., 1991). Corticospinal projections in marsupials terminate in the dorsal and intermediate parts of the spinal gray matter, predominantly lamina IV and medial lamina V and VI (Martin and Fisher, 1968; Martin et al., 1970; Rees and Hore, 1970; Martin et al., 1975; Holst et al., 1991). In contrast, corticospinal axons in rats and monkeys are

distributed in the entire length of the cord and terminate in lower lamina including lamina IX, the lamina containing motoneuronal cell bodies of the ventral horn (Valverde, 1966; Brown, 1971; Kuypers, 1981; Roullier et al., 1991; Bortoff and Strick, 1993). Differences in both the number of corticospinal neurons and in their final termination sites in opossums and other mammalian species may contribute to the differences in the ability to evoke body movements upon microstimulation of cortex. These anatomical differences in the corticospinal tract may also contribute to differences observed in behavioral tests of dexterity in different mammalian species. For example, rats display more complex manipulations of prey items and more complex reaching behavior compared to more stereotyped movements of opossums (Ivanco et al., 1996). Ivanco et al. (1996) found that *Monodelphis* was more likely to use the mouth for manipulation of prey, suggesting that corticobulbar control of the face may have developed early in cortical evolution for more complex head movements. Furthermore, ablation behavior studies of motor cortex in rats have shown that transection of the pyramidal tract (Whishaw et al., 1993) and lesions of motor cortex (Whishaw et al., 1992) result in simplified reaching strategies resembling those displayed by *Monodelphis.*

Evolution of sensorimotor cortex

Lende's view that S1 of *Didelphis* is an "amalgam" of S1 and M1 suggests that separate S1 and M1 areas evolved in placental mammals from a single area possessing both sensory and motor functions. More recent studies suggest that motor responses evoked upon stimulation of S1 are similar to the higher threshold S1 motor responses commonly seen in other mammals (Beck et al., 1996). Beck et al. (1996) found that even the few sites rostral to the sensory representation where movements were evoked required higher thresholds than one would expect in an M1 representation in eutherian mammals (Neafsey et al., 1986; Sessle and Weisendanger, 1982; Stepniewska et al., 1993). These results suggest that the amalgam proposed by Lende (1963a) might have been a reflection of motor components within the S1 representation, similar to that found in the S1 of other mammals. The electrophysiological results presented here in *Monodelphis* suggest that this species has a motor component within S1, though it is not as well-developed (i.e., complete) as that observed in other mammals, and not as developed as that seen in *Didelphis.* This result may represent a primordial S1 condition, in which a complete somatosensory map has been achieved and retained, but the motor components of S1 have not yet fully developed. In this primitive condition, the anatomical substrate required for cortical control of movements below the level of the face is not present.

The cytoarchitecture of the vibrissae motor response area in S1 of *Monodelphis* also suggests that this is a sensory area rather than a true motor cortex. The vibrissae response area has a notable granular layer (layer IV) and does not have the agranular appearance notable of true motor cortex (Fig. 15; and see Gray, 1924; Walsh and Ebner, 1970 in *Didelphis*). In addition, the myelin staining pattern is similar to that shown for somatosensory cortex of other mammals (see Krubitzer, 1995).

Sensorimotor cortex in *Monodelphis* appears to be the least developed in any mammal studied thus far, and may represent the primordial condition from which primate sensory and motor cortex arose. Figure 17 illustrates a hypothetical model of evolutionary divergence of sensorimotor cortex supported by the results reported here. *Monodelphis* has a complete S1 representation, as well as a second, S2 representation, with an *incomplete* motor component overlapping the S1 representation of the face (grade 1). *Didelphis* has a complete S1 representation (as well as several additional representations (Beck et al., 1996) with a *complete* motor representation overlapping the somatosensory representation in S1 (grade 2; Lende, 1963a). This S1 motor representation in *Didelphis* is not unlike that found in S1 of placental mammals, as high-threshold stimulation is required to evoke movement. A true motor cortex, as evidenced by a reversed motor representation rostral to the somatosensory representation and movements evoked at relatively low stimulating currents, is apparent in rats (grade 3). In this grade, the separation of S1 and M1 is not complete, because an overlap has been demonstrated over most of the hindlimb representation and part of the forelimb representation (Hall and Lindholm, 1974; Donoghue and Wise, 1982; Sanderson et al., 1983). A complete and separate motor cortex, as evidenced by a reversed motor representation rostral to the somatosensory representation and low-threshold evoked movement, is apparent in primates (grade 4).

The cortical organization in monotremes, however, suggests that sensorimotor differentiation may not have followed this simple linear progression. For example, by using surface stimulation techniques in the prototherian echidna, Lende (1964) found a complete motor representation overlapping the somatosensory representation. By using surface stimulation techniques in the platypus, Bohringer and Rowe (1977) found a partial overlap of the motor and somatosensory representations. Electrical stimulation evoked movements of the bill and forelimb, but not the hindlimb. Thus, it would appear that the motor component of somatosensory cortex is more complete in monotremes than in *Monodelphis* opossums.

Furthermore, an area rostral to somatosensory cortex in monotremes is in some ways characteristic of motor cortex. In early surface stimulation studies in echidna (Abbie, 1938; Goldby, 1939), evoked movements were observed in an area rostral to sulcus α, and thus rostral to S1 (Krubitzer et al., 1995). Surface stimulation studies in platypus also resulted in evoked movements rostral to S1, including representations of the forelimb. Based on electrophysiogical and cytoarchitectonic criteria, two fields rostral to S1 have been identified in both echidna and platypus (Krubitzer et al., 1995). The rostral field (field R) contains neurons responsive to stimulation of deep receptors, whereas the manipulation field (field M) has the architectonic appearance of the eutherian M1 (Ulinski, 1984; Krubitzer et al., 1995).

These findings of a more complete motor component in somatosensory cortex and a possible M1 homologue rostral to S1 suggest that a more developed sensorimotor area may have existed in mammals previous to the emergence of Didelphids. If so, then sensorimotor cortex in *Monodelphis* may be the result of secondary simplification. Alternatively, the forelimb component of S1 in monotremes may be homoplaseous to that of *Didelphis,* having arisen through convergent evolution. Similarly, the rostral motor area in monotremes may be homoplaseous to M1 in eutherian mammals.

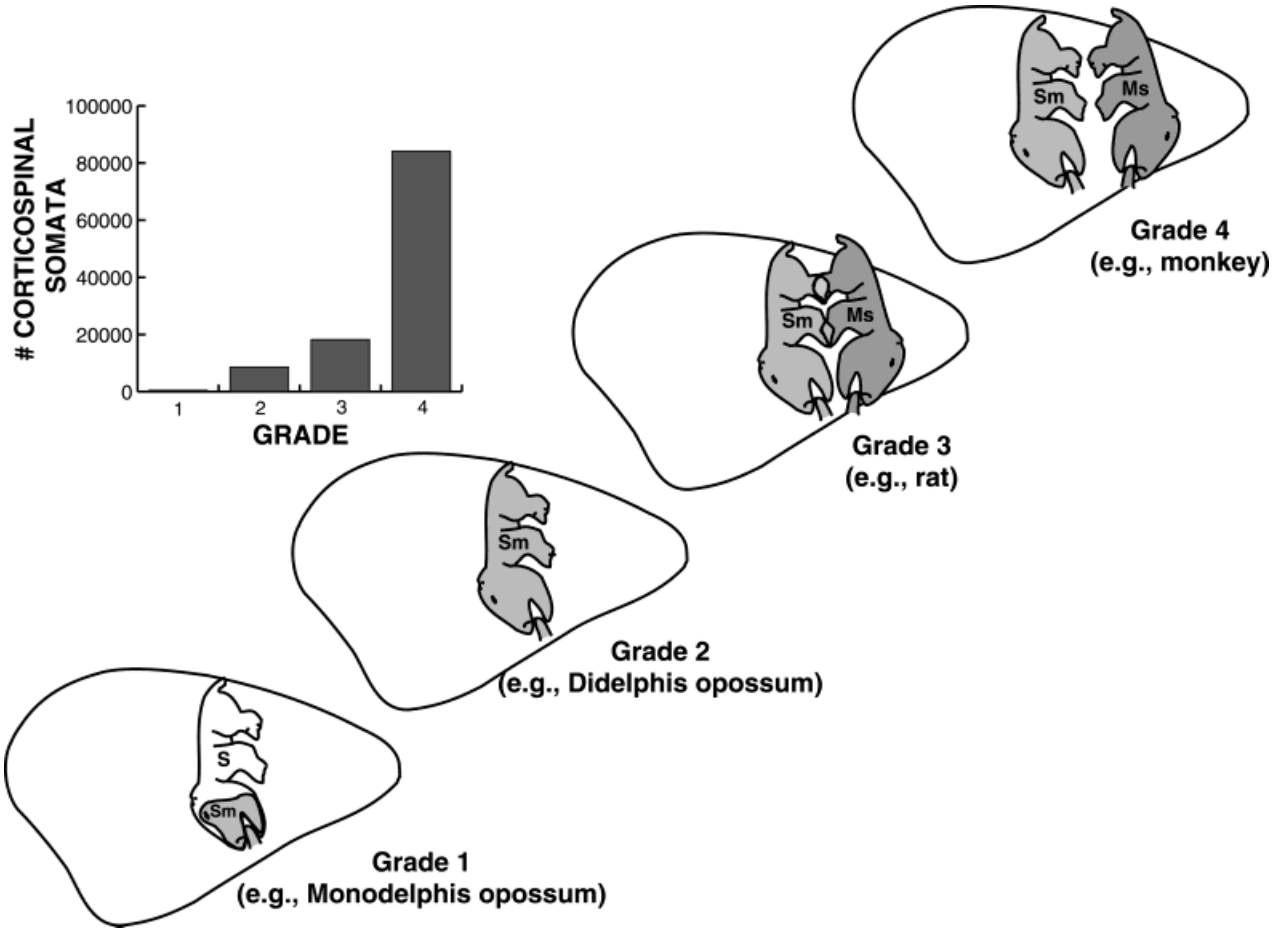


Fig. 17. Hypothetical model of evolutionary divergence in sensorimotor cortex. Grade 1, a complete somatosensory (S) representation exists. An incomplete motor representation is congruent with the sensory representation of the face (Sm). Grade 2, a complete motor representation exists and overlaps the somatosensory (Sm). As noted by Lende (1963a), this "amalgam" is not unlike primary somatosensory area (S1) of placental mammals. A reversed map characteristic of true motor cortex is absent. Grade 3, true motor cortex has emerged, as evident from a reversed motor representation (Ms) rostral to the somatosensory representation (Sm). Ms is largely segregated from Sm, though a partial overlap exists in the hindlimb and forelimb areas (e.g., see Hall and Lindholm, 1974; Donoghue and Wise, 1982). Grade 4, Ms is completely segregated from Sm. Somatosensory cortex retains some motor properties, whereas motor cortex retains some somatosensory properties. Upper left, the increase in the number of corticospinal neurons parallels the evolutionary divergence of sensory and motor areas (Nudo and Masterton, 1990). Figure 17 is in Adobe Illustrator 7.0.1.

Although the interpretation of these results in monotremes remains speculative, it is now clear that the opossum's "sensorimotor amalgam" is *not* motor cortex in the same sense as it is in grade 3 and 4 mammals. The somatosensory cortex of grade 3 and 4 mammals is not without some motor properties, as mentioned above. But a motor component in the opossum's sensorimotor area is not, by itself, sufficient grounds to consider it motor cortex. Correspondence of sensory and motor maps in the opossum's sensorimotor area is just what is seen in grade 3 and 4 mammals. Therefore, the sensorimotor "amalgam" of opossum cortex is probably no more "motor" than the sensory cortex of, for example, monkey. In addition to the differentiation of somatosensory fields, as has been reported in a number of recent papers, it would appear that the motor component of this predominantly somatosensory cortical field developed in stages. In addition, it is probably safe to suggest that in opossum cortex specifically and in primitive, or grade one and two cortex generally, there is no true motor cortex. If this is the case (and opossums provide a reasonable representation of the primitive conditions), then motor cortex must have arisen anew sometime later in evolution. It is not entirely clear when true motor cortex might have emerged. This riddle may be difficult to solve, because representatives of intermediate conditions may not exist among extant mammals. However, as evolutionary pressures for greater dexterity of the distal skeletal musculature grew, augmentations of the corticospinal tract from a primitive to a more advanced pathway seem to have paralleled the origin of a true motor cortex.

ACKNOWLEDGMENTS

The authors thank Terry Allen for help in data collection and Anna Walton-Williams for help in the histological preparations.

LITERATURE CITED

Abbie AA. 1938. The excitable cortex in Monotremata. Aust J Exp Biol Med Sci 16:143–152.

Aizawa H, Mushiake H, Inase M, Tanji J. 1990. An output zone of the monkey primary motor cortex specialized for bilateral hand movement. Exp Brain Res 82:219–221.

Asanuma H, Arissian K. 1984. Experiments on functional role of peripheral input to motor cortex during voluntary movements in the monkey. J Neurophysiol 52:212–227.

Asanuma H, Larsen KD, Yumiya H. 1979. Direct sensory pathway to the motor cortex: a basis of cortical reflexes. In: Asanuma H, Wilson VJ, editors. Integration in the nervous system. Tokyo/New York: Igakushoin. p 223–238.

Asanuma H, Larsen KD, Yumiya H. 1980. Peripheral input pathways to the monkey motor cortex. Exp Brain Res 38:349–355.

Asanuma H, Stoney SD, Jr, Abzug C. 1968. Relationship between afferent input and motor outflow in cat motorsensory cortex. J Neurophysiol 31:670–681.

Beck PD, Pospichal MW, Kaas JH. 1996. Topography, architecture, and connections of somatosensory cortex in opossums: evidence for five somatosensory areas. J Comp Neurol 366:109–133.

Bermejo R, Harvey M, Gao P, Philip Zeigler H. 1996. Conditioned whisking in the rat. Somatosens Mot Res 13:225–233.

Bohringer RC, Rowe MJ. 1977. The organization of the sensory and motor areas of cerebral cortex in the platypus (*Ornithorhynchus anatinus*). J Comp Neurol 174:1–14.

Bortoff GA, Strick PL. 1993. Corticospinal terminations in two new-world primates: further evidence that corticomotoneuronal connections provide part of the neural substrate for manual dexterity. J Neurosci 13:5105–5118.

Brecht M, Preilowski B, Merzenich MM. 1997. Functional architecture of the mystacial vibrissae. Brain Res Behav Brain Res 84:81–97.

Brodmann K. 1909. Vergleichende lokalisationslehre der grosshirnrinde. Leipzig: Verlag Barth.

Brown LT. 1971. Projections and termination of the corticospinal tract in rodents. Exp Brain Res 13:432–450.

Carvell GE, Simons DJ. 1990. Biometric analysis of vibrissal tactile discrimination in the rat. J Neurosci 10:2638–2648.

Carvell GE, Simons DJ. 1995. Task- and subject-related differences in sensorimotor behavior during active touch. Somatosens Mot Res 12: 1–9.

Catania KC, Kaas J. 1995. organization of the somatosensory cortex of the star-nosed mole. J Comp Neurol 351:549–567.

Catania K, Lyon D, Mock O, Kaas J. 1999. Cortical organization in shrews: evidence from 5 species. J Comp Neurol 410:55–72.

Clemens WA. 1968. Origin and early evolution of marsupials. Evolution 22:1–18.

Clemens WA. 1977. Phylogeny of marsupials. In: Stonehouse B, Gilmore D, editors. The biology of marsupials. London: Macmillan. p 51–68.

Clemens WA. 1989. Diagnosis of the class Mammalia. In: Walton DW, Richardson BJ, editors. Fauna of Australia, vol. 1B, mammalia. Canberra: Australian Government Publishing Services. p 401–406.

Coleman GT, Zhang HQ, Murray GM, Zachariah MK, Rowe MJ. 1999. Organization of somatosensory areas I and II in marsupial cerebral cortex: parallel processing in the possum sensory cortex. J Neurophysiol 81:2316–2324.

Cramer S, Finklestein SP, Schaecter JD, Bush G, Rosen BR. 1999. Activation of distinct motor cortex regions during ipsilateral and contralateral finger movements. J Neurophysiol 81:383–387.

Doetsch GS, Gardner EP. 1972. Relationship between afferent input and motor output in sensorimotor cortex of the monkey. Exp Neurol 35:78–97.

Donoghue JP, Wise SP. 1982. The motor cortex of the rat: cytoarchitecture and microstimulation mapping. J Comp Neurol 212:76–88.

Dykes RW, Rasmusson DD, Hoeltzell P. 1980. Organization of primary somatosensory cortex in the cat. J Neurophysiol 43:1527–1546.

Edinger T. 1948. Evolution of the horse brain. Mem Geol Soc Amer 25:1–177.

Elston GL, Krubitzer L, Manger P, Calford M, Day T. 1993. The organization and connections of somatosensory cortex in the Australian marsupial, brush-tailed possum (*Trichosurus vulpecula*). Soc Neurosci Abstr 19:764.

Fadem BH, Cole E. 1985. Scent marking in the gray short-tailed opossum (*Monodelphis domestica*). Anim Behav 33:730–738.

Fadem BH, Corbett A. 1993. Sex differences and the role of aromatization in the control of sexually dimorphic behavior and morphology in gray short-tailed opossums (*Monodelphis domestica*). Horm Behav 27:366–379.

Felleman DJ, Wall JT, Cusick CG, Kaas JH. 1981. Evidence for a single cutaneous representation within the SI region of the cat. Anat Rec 199:82a.

Flannery TF. 1989. Origins of the Australo-Pacific fauna. Aust Zool Rev 1:15–24.

Frost SB, Masterton RB. 1992. Origin of auditory cortex. In: Webster DB, Fay RR, Popper AN, editors. The evolutionary biology of hearing. New York: Springer-Verlag. p 655–671.

Gallayas F. 1979. Silver staining of myelin by means of physical development. Neurology 1:203–209.

Gioanni Y, Lamarche M. 1985. A reappraisal of rat motor cortex organization by intracortical microstimulation. Brain Res 8:177–194.

Goldby F. 1939. An experimental investigation of the motor cortex and pyramidal tract of *Echidna aculeata.* J Anat 73:509–524.

Gould HJ, Cusick CG, Pons TP, Kaas JH. 1986. The relationship of corpus collosum connections to electrical stimulation maps of motor, supplementary motor, and frontal eye fields in owl monkeys. J Comp Neurol 247:297–325.

Gray PA. 1924. The cortical lamination of the opossum, *Didelphys virginiana.* J Comp Neurol 37:221–263.

Gregory WK. 1929. Our face from fish to man. Putnam's Sons, reprinted in 1965 by Capricorn, New York.

Hall RD, Lindholm EP. 1974. Organization of motor and somatosensory neocortex in the albino rat. Brain Res 66:23–38.

Holst MC, Ho RH, Martin GF. 1991. The origins of supraspinal projections to lumbosacral and cervical levels of the spinal cord in the gray short-tailed Brazilian opossum, *Monodelphis domestica.* Brain Behav Evol 38:273–289.

Huffman KJ, Nelson J, Clarey J, Krubitzer L. 1999. Organization of somatosensory cortex in three species of marsupials, *Dasyurus hallucatus, Dactylopsila trivirgata,* and *Monodelphis domestica:* neural correlates of morphological specializations. J Comp Neurol 403:5–32.

Hutson KA, Masterton RB. 1986. The sensory contribution of a single vibrissa's cortical barrel. J Neurophysiol 56:1196–1223.

Ivanco TL, Pellis SM, Wishaw IQ. 1996. Skilled forelimb movements in prey catching and reaching by rats (*Rattus norvegicus*) and opossums (*Monodelphis domestica*): relations to anatomical differences in motor systems. Brain Res Behav Brain Res 79:163–181.

Johnson JI. 1990. Comparative development of somatic sensory cortex. In: Jones EG, Peters A, editors. Cerebral cortex. New York: Plenum. p 335–449.

Johnson JI, Haight JR, Migirian D. 1973. Convolutions related to sensory projections in cerebral cortex of marsupial wombats. J Anat 114:153.

Kaas JH. 1983. What if anything is S1? Organization of first somatosensory area of cortex. Physiol Rev 63:206–230.

Kaas JH. 1987. The organization and evolution of neocortex. In: Wise SP, editor. Higher brain functions. New York: John Wiley and Sons. p 347–378.

Kaas JH. 1989. The evolution of complex sensory systems in mammals. J Exp Biol 146:165–176.

Kimble DP. 1997. Didelphid behavior. Neurosci Biobehav Rev 21:361–369.

Kirsch JAW. 1977. The six-percent solution: second thoughts on the adaptedness of marsupiala. Am Scientist 65:276–288.

Kleinfeld D, Berg RW, O'Connor SM. 1999. Anatomical loops and their electrical dynamics in relation to whisking by rat. Somatosens Mot Res 16:69–88.

Krishnamurti A, Sanides F, Welker WI. 1976. Microelectrode mapping of modality-specific somatic sensory cerebral neocortex in slow loris. Brain Behav Evol 13:367–383.

Krubitzer L. 1995. The organization of neocortex in mammals: are species differences really so different? Trends Neurosci 18:408–417.

Krubitzer LA. 1996. The organization of lateral somatosensory cortex in primates and other mammals. In: Franzen O, Johansson R, Terenius L, editors. Somesthesis and the neurobiology of the somatosensory cortex. Basel: Birkhauser Verlag. p 173–185.

Krubitzer LA, Kaas JH. 1990. The organization and connections of somatosensory cortex in marmosets. J Neurosci 10:952–974.

Krubitzer LA, Sesma MA, Kaas JH. 1986. Microelectrode maps, myeloar-

chitecture, and cortical connections of three somatotopically organized representations of the body surface in the parietal cortex of squirrels. J Comp Neurol 250:403–430.

Krubitzer LA, Calford MB, Schmid LM. 1993. Connections of somatosensory cortex in megachiropteran bats: the evolution of cortical fields in mammals. J Comp Neurol 327:473–506.

Krubitzer LA, Manger P, Pettigrew J, Calford M. 1995. Organization of somatosensory cortex in monotremes: in search of the prototypical plan. J Comp Neurol 351:261–306.

Krubitzer L, Kunzle H, Kaas J. 1997. Organization of sensory cortex in a madagascan insectivbore, the tenrec (*Echinops telfairi*) J Comp Neurol 379:399–414.

Kuypers HGJM. 1981. Anatomy of the descending pathways. In: Brookhart JM, Mountcastle VB, editors. Handbook of physiology: the nervous system, vol 2, part 1. Baltimore, MD: Williams and Wilkins. p 597–666.

Lende RA. 1963a. Cerebral cortex: a sensorimotor amalgam in the marsupiala. Science 141:730–732.

Lende RA. 1963b. Sensory representation in the cerebral cortex of the opossum (*Didelphis virginiana*). J Comp Neurol 121:395–403.

Lende RA. 1963c. Motor representation in the cerebral cortex of the opossum (*Didelphis virginiana*). J Comp Neurol 121:405–415.

Lende RA. 1964. Representation in the cerebral cortex of a primitive mammal. J Neurophysiol 27:37–48.

Loo YT. 1930. The forebrain of the opossum. J Comp Neurol 51:13–64.

Magalhåes-Castro B, Saraiva PES. 1972. Sensory and motor representation in the cerebral cortex of the marsupial *Didelphis azarae azara.* Brain Res 34:291–299.

Malis LI, Pribram KH, Kruger L. 1953. Action potentials in motor cortex evoked by peripheral nerve stimulation. J Neurophysiol 16:161–167.

Marshall LG. 1979. Evolution of metatherian and eutherian (mammalian) characters: a review based on cladistic methodology. Zool J Linnean Soc 66:369–410.

Marshall LG, Case JA, Woodburne MD. 1987. Phylogenetic relationships of the families of marsupials. In: Genoways HH, editor. Current mammalogy. New York: Plenum Press. p 433–481.

Martin GF, Beattie MS, Bresnahan JC, Henkel CK, Hughes HC. 1975. Cortical and brain stem projections to the spinal cord of the American opossum (*Didelphis marsupialis virginiana*). Brain Behav Evol 12: 270–310.

Martin GF, Fisher AM. 1968. A further evaluation of the origin, course and termination of the opossum corticospinal tract. J Neurol Sci 7:177–189.

Martin GF, Megirian D, Roebuck A. 1970. The corticospinal tract of the marsupial phalanger (*Trichosurus vulpecula*) J Comp Neurol 139:245–258.

Melzer P, Hackett T, Masterton RB, Kaas JH. 1997. Mapping of auditory cortex in the mouse opossum, *Monodelphis domestica.* Soc Neurosci Abstr 23:2072.

Merzenich MM, Kaas JH, Sur M, Lin CS. 1978. Double representation of the body surface within cytoarchitectonic areas 3b and 1 in "S1" in the owl monkey (*Aotus trivirgatus*). J Comp Neurol 181:41–74.

Neafsey EJ, Bold EL, Haas G, Hurley-Gius KM, Quirk G, Sievert CF, Terreberry RR. 1986. The organization of the rat motor cortex: a microstimulation mapping study. Brain Res 11:77–96.

Nudo RJ, Masterton RB. 1988. Descending pathways to the spinal cord: HRP study in 22 mammals. J Comp Neurol 277:53–79.

Nudo RJ, Masterton RB. 1990a. Descending pathways to the spinal cord, III: Sites of origin of the corticospinal tract. J Comp Neurol 296:559–583.

Nudo RJ, Masterton RB. 1990b. Descending pathways to the spinal cord, IV: some factors related to the amount of cortex devoted to the corticospinal tract. J Comp Neurol 296:584–597.

Nudo RJ, Jenkins WM, Merzenich MM, Prejean T, Grenda R. 1992. Neurophysiological correlates of hand preference in primary motor cortex of adult squirrel monkeys. J Neurosci 12:2918–2947.

Nudo RJ, Sutherland DP, Masterton RB. 1995. Variation and evolution of mammalian corticospinal somata with special reference to primates. J Comp Neurol 358:181–205.

Nudo RJ, Milliken GW, Jenkins WM, Merzenich MM. 1996. Use-dependent alterations of movement representations in primary motor cortex of adult squirrel monkeys. J Neurosci 16:785–807.

Nudo RJ, Plautz EJ, Milliken GW. 1997. Adaptive plasticity in primate motor cortex as a consequence of behavioral experience and neuronal injury. Semin Neurosci 9:13–23.

Olsen EC. 1959. The evolution of mammalian characters. Evolution 13: 344–353.

Penfield W, Boldrey E. 1937. Somatic motor and sensory representation in the cerebral cortex of man as studied by electrical stimulation. Brain 60:389–443.

Penfield W, Rasmussen T. 1952. The cerebral cortex of man. New York: Macmillan.

Pobirsky N, Molnar Z, Blakemore C, Krubitzer L. 1998. The organization of somatosensory cortex in the west European hedgehog (*Erinaceus europaeus*). Soc Neurosci Abstr 24:1125.

Pocock RI. 1914. On the facial vibrissae in the mammalia. Proc Zool Soc Lond (1914) 889–912.

Poran NS, Tripoli R, Halpern M. 1993a. Nuzzling in the gray short-tailed opossum I: delivery of odors to vomeronasal organ. Physiol Behav 53:959–967.

Poran NS, Tripoli R, Halpern M. 1993b. Nuzzling in the gray short-tailed opossum II: familiarity and individual recognition. Physiol Behav 53: 969–973.

Pubols BH. 1977. The second somatic sensory area (SmII) of opossum neocortex. J Comp Neurol 174:71–78.

Pubols BH, Pubols LM, DiPette DJ, Sheely JC. 1976. Opossum somatic sensory cortex: a microelectrode mapping study. J Comp Neurol 165: 229–246.

Rees S, Hore J. 1970. The motor cortex of the brush-tailed possum (*Trichosurus vulpecula*): motor representation, motor function and the pyramidal tract. Brain Res 20:439–451.

Reig OA, Kirsch JAW, Marshall LG. 1987. Relationships of American "opossum-like" marsupials. In: Archer M, editor. Possums and opossums: studies in evolution, vol. I. Australia: Surry Beatty and Sons Limited. p 10–89.

Rosén I, Asanuma H. 1972. Peripheral afferent inputs to the forelimb area of the monkey motor cortex: input-output relations. Exp Brain Res 14:257–273.

Roullier EM, Liang F, Moret V, Wiesendanger M. 1991. trajectory of redirected corticospinal axons after unilateral lesion of the sensorimotor cortex in neonatal rat: a *Phaseolus vulgarus leucoagglutinin* (PHA-L) tracing study. Exp Neurol 114:53–65.

Rowe M. 1990. Organization of the cerebral cortex in monotremes and marsupials. In: Jones EG, Peters A, editors. Cerebral cortex. New York: Plenum Press. p 253–334.

Sanderson KJ, Welker W, Shambes GM. 1983. Reevaluation of motor cortex and sensorimotor overlap in cerebral cortex of albino rats. Brain Res 292:251–260.

Sapienza S, Talbi B, Jacquemin J, Albe-Fessard D. 1981. Relationship between input and output of cells in motor and somatosensory cortices of the chronic awake rat. Exp Brain Res 43:47–56.

Schafer EA. 1900. Textbook of physiology, vol 2, the cerebral cortex. London: Young J. Pentland. p 697–782.

Sessle BJ, Wiesendanger M. 1982. Structural and functional definition of the motor cortex in the monkey (*Macaca fascicularis*). J Physiol (Lond) 323:245–265.

Simpson GG. 1949. The meaning of evolution. New Haven, CT: Yale University Press.

Simpson GG. 1959. Mesozoic mammals and the polyphyletic origin of mammals. Evolution 13:405–414.

Smith GE. 1910. Some problem relating to the evolution of the brain. Lancet 1:1–6, 147–153, 221–227.

Stepniewska I, Preuss TM, Kaas JH. 1993. Architectonics, somatotopic organization, and ipsilateral cortical connections of the primary motor area (M1) of owl monkeys. J Comp Neurol 177:569–588.

Strick PL, Preston JB. 1982. Two representations of the hand in area 4 of a primate. II. Somatosensory input organization. J Neurophysiol 48: 150–159.

Strilein KE. 1982. Behavior, ecology and distribution of South American marsupials. In: Mores MA, Genoways HH, editors. Mammalian biology in South America. Linesville, PA: Pymatuning Laboratory of Ecology. p 231–249.

Sur M, Nelson RJ, Kaas JH. 1980a. The representation of the body surface in somatic koniocortex in the prosimian *Galago.* J Comp Neurol 189: 381–402.

Sur M, Weller RE, Kaas JH. 1980b. Representation of the body surface in somatosensory area 1 of tree shrews, *Tupaia glis.* J Comp Neurol 194:71–95.

Sur M, Nelson RJ, Kaas JH. 1982. Representations of the body surface in

cortical areas 3b and 1 of squirrel monkeys: comparisons to other primates. J Comp Neurol 211:177–192.

Tanji J, Wise SP. 1981. Submodality distribution in sensorimotor cortex of the unanesthetized monkey. J Neurophysiol 45:467–481.

Thompson D, Stoney SD, Asanuma H. 1970. Characteristics of projections from primary sensory cortex to motor-sensory cortex in cats. Brain Res 22:15–27.

Ulinski PS. 1984. Thalamic projections to the somatosensory cortex of the echidna, *Tachyglossus aculeatus.* J Comp Neurol 229:153–170.

Valverde F. 1966. The pyramidal tract in rodents. A study of its relations with the posterior column nuclei, dorsolateral reticular formation of the medulla oblongata, and cervical spinal cord (Golgi and electron microscopic observations). Z Zellforsch 71:297–363.

von Economo C. 1929. The cytoarchitectonics of the human cortex. Oxford: Oxford University Press.

Walsh TM, Ebner FF. 1970. The cytoarchitecture of somatic sensory motor cortex in the opossum (*Didelphis marsupialis virginiana*): A Golgi study. J Anat 107:1–18.

Welker WI. 1964. Analysis of sniffing of the albino rat. Behavior 22:223–244.

Welker WI, Benjamin RM, Miles RC, Woolsey CN. 1957. Motor effects of cortical stimulation in squirrel monkey (*Saimiri sciureus*). J Neurophysiol 20:347–364.

Weller WL. 1993. SM1 Cortical barrels in an Australian marsupial, *Trichosurus vulpecula* (brush-tailed opossum): structural organization, patterned distribution, and somatotopic relationships. J Comp Neurol 337:471–492.

Welt C, Aschoff J, Kameda K, Brooks VB. 1967. Intracortical organization of cat's sensory motor neurons. In: Purpura DP, Yahr MD, editors. Neurophysiological basis of normal and abnormal motor activities. Hewlett, NY: Raven Press p 255–293.

Westerman M, Edwards D. 1992. DNA hybridization and the phylogeny of monotremes. In: Augee ML, editor. Platypus and echidnas. Sydney: The Royal Zoological Society of New South Wales. p 28–34.

Whishaw IQ, Pellis SM, Gorny BP. 1992. Medial frontal cortex lesions impair the aiming component of rat reaching. Behav Brain Res 50:93–104.

Whishaw IQ, Pellis SM, Gorny BP, Kolb B, Tetzlaff W. 1993. Proximal and distal impairments in rat forelimb use in reaching follow unilateral pyramidal tract lesions. Behav Brain Res 56:59–76.

Wiesendanger M. 1973. Input from muscle and cutaneous nerves of the hand and forearm to neurones of the precentral gyrus of baboons and monkeys. J Physiol (Lond) 228:203–219.

Woolsey CN. 1947. Patterns of sensory representation in the cerebral cortex. Fed Proc 6:437–464.

Woolsey CN. 1958. Organization of somatic sensory and motor areas of the cerebral cortex. In: Harlow HF, Woolsey CN, editors. Biological and biochemical bases of behavior. Madison: University of Wisconsin Press. p 63–82.

Woolsey CN, Travis AM, Barnard JW, Ostenso RS. 1953. Motor representation in the postcentral gyrus after chronic ablation of precentral and supplementary motor areas. Fed Proc 12:160.